SIMATS ENGINEERING

Department of Information Technology

ITA05 – COMPUTER VISION

ASSIGNMENT 1 – CO1

Apply, Compare and Justify Image Acquisition, Formation and Processing Techniques for a Vision-Based E-Waste Battery Sorting Line

CO1: Apply the fundamental concepts, image formation models, and processing techniques for analyzing Computer Vision systems.

Student Details	**Information**
Student Name	KESHAVARAM L
Register Number	192511253
Department / Section	Computer Science and Engineering
Academic Year / Semester	2025–2026 / MS2
Faculty Name	Dr Murali P
Nature of Assignment	Individual
Duration	1 Week
Total Marks	50
Date of Submission	31.10.2026
Date of Evaluation	____.____._____

Scenario: Vision-Based E-Waste Battery Sorting Line

GreenCycle Robotics Corp. requires an automated vision system to classify cylindrical 18650, cylindrical 21700 and prismatic pouch batteries; read LCO, LFP, NMC and NiMH chemistry codes; and detect rust, dents and pouch swelling before a pneumatic reject gate.

The design uses the supplied operating constraints: 320 mm × 240 mm FOV, 1.2 m/s belt speed, up to 15 cells/s, 400 mm camera height, 400 ms decision window, 0.3 mm minimum defect, 0.4 mm text stroke and a maximum 1.5% miss/misclassification target.

MARK SHEET (Rubric-Based Evaluation)

Task	Task Focus	Maximum Marks	Marks Awarded	Remarks
1	Systems comparison: manual, assisted and automated sorting	4	_____	
2	Pipeline level depth comparison	5	_____	
3	Color-space representation for rust analysis	4	_____	
4	Pixel connectivity and scratch counting	4	_____	
5	Sensor shutter and exposure topology	5	_____	
6	Sampling and spatial resolution	5	_____	
7	Brightness quantization and dynamic range	4	_____	
8	Geometric projection and photometric correction	5	_____	
9	Integrated front-end optimization	6	_____	
10	High-speed plant scale-up challenge	8	_____	
TOTAL	Comprehensive evaluation score	50	_____	

OVERALL REMARKS

_____

_____

Student Signature: KESHAVARAM L

Faculty Signature: _____

Program Director: _____

MANDATORY SUBMISSION EVIDENCE CHECK

Task	Evidence Expected	Status	Note
1	Comparison table, speed calculation, agreement/objectivity discussion and justified choice	☐	
2	Level-annotated pipeline, accuracy/latency comparison and shadow failure analysis	☐	
3	RGB/HSV/Lab comparison, illumination analysis, array layout and overflow proof	☐	
4	Connectivity code, synthetic patterns, component counts and distance comparison	☐	
5	Blur calculations, rolling/global/line-scan comparison and signal chain	☐	
6	Resolution, mm/px, defect samples, bandwidth, storage, Nyquist and frame-rate calculations	☐	
7	Bit-depth comparison, contrast evidence, storage and model quantization distinction	☐	
8	Focal length, perspective error, calibration and flat-field correction	☐	
9	Three packages, budget, end-to-end diagram and deadline budget	☐	
10	Scale-up calculations, strategy comparison and RETAIN/MODIFY/REDESIGN decision	☐	

EXPERIMENTAL SETUP AND ENGINEERING ASSUMPTIONS

The assignment gives the operating constraints but does not provide real battery photographs, measured camera timings, supplier quotations or a ground-truth inspection dataset. Therefore, mathematical values below are calculated from the supplied specifications. Where accuracy or latency values are used for design comparison, they are explicitly labelled as engineering estimates or target values, not as measured experimental results.

Parameter	Given value / design interpretation
Conveyor speed	1.2 m/s = 1,200 mm/s
Field of view	320 mm across × 240 mm along belt
Camera height	400 mm above belt; vertical overhead view
Throughput	15 cells/s; average arrival interval = 1/15 = 66.7 ms
Rejector distance	0.48 m downstream; travel time = 0.48/1.2 = 0.400 s = 400 ms
Smallest defect	0.3 mm micro-crack / rust pit
Text stroke	0.4 mm
Pouch dimensional tolerance	±0.5 mm
Lighting variation	1,500–45,000 lux; 100 Hz flicker from 50 Hz mains
Compute	4 CPU cores, 16 GB RAM, no GPU; Python, OpenCV, NumPy
Storage / budget	1.5 TB SSD; hardware budget ₹1,25,000
Acceptance limit	Misclassification / defect miss rate ≤1.5%

Common calculations

Reject timing: t = distance / speed = 0.48 m / 1.2 m s^{-1} = 0.400 s = 400 ms. This is the complete camera-to-piston time budget; image processing must finish early enough to communicate the decision and actuate the reject mechanism.

Cell arrival interval: 1 / 15 = 0.0667 s = 66.7 ms. The system must therefore accept a new cell at least every 66.7 ms even though the same cell has a 400 ms travel window to the rejector.

For the calculations, decimal MB and TB are used: 1 MB = 10^6 bytes and 1 TB = 10^{12} bytes. Image-rate storage assumes one full uncompressed RGB image per cell at 15 images/s, unless stated otherwise.

System-level design objective

The front-end should produce repeatable, glare-controlled, geometrically calibrated images. A practical choice is a global-shutter area-scan camera, short synchronized LED strobe, diffuse dome, calibrated low-distortion lens and a three-level classical vision pipeline. The final design must still be validated on real battery images before claiming the required ≤1.5% error rate.

TASK 1 – Systems Comparison: Manual, Assisted and Automated Sorting [4 Marks]

The line has to inspect up to 15 batteries every second. The three techniques differ in who makes the final decision and how consistently that decision can be repeated.

1.1 Comparison of the three workflows

Criterion	A. Manual inspection	B. Image-processing-assisted	C. Automated computer vision
Processing speed	A trained person may inspect roughly 1–3 cells/s depending on defect complexity; several operators would be needed.	Enhancement is fast, but the operator remains the bottleneck; usually similar to manual inspection.	Designed for ≥15 cells/s; pipeline must process each arriving cell within 66.7 ms on average.
Repeatability	Fatigue, attention and experience change decisions; intra-rater agreement can fall during long shifts.	Zoom and contrast help, but human judgement still varies.	Same calibrated rules and thresholds are applied to every cell; deterministic repeatability.
Objectivity / Cohen's kappa	Inter-rater κ may be low or moderate because operators interpret defects differently.	May improve agreement, but κ still depends on operators.	No inter-rater variation; compare output against ground truth using accuracy, precision, recall and confusion matrix. κ=1 between identical repeated deterministic runs is not proof of accuracy.
Non-contact safety	Cells may be handled or rotated, increasing risk of contact with damaged cells.	Camera inspection is non-contact, but manual handling may remain.	Fully non-contact inspection; hazardous swollen or damaged cells can be rejected without handling.
Continuous	Limited by fatigue,	Operator fatigue remains.	Can run continuously with

endurance	breaks and shift length.		planned maintenance and thermal management.
Traceability	Paper records may be incomplete.	Images may be saved, but decisions can be inconsistently recorded.	Store image, timestamp, cell ID, OCR result, measurements, defect flags and reject decision.
Lifecycle cost	Recurring labour and training costs.	Labour plus camera, display and lighting costs.	Higher initial hardware/software cost; lower repetitive labour and stronger traceability.
15 cells/s suitability	Not suitable for a single operator; requires parallel stations.	Not suitable if every frame needs human approval.	Best fit, provided measured end-to-end throughput and accuracy meet requirements.

1.2 Throughput and agreement

At 15 cells/s, one cell arrives every 66.7 ms. A single human taking even 0.5 s per cell can inspect only 2 cells/s. The shortfall is 13 cells/s, so at least eight such operators would be required in parallel before allowing for breaks, hand-offs and disagreement. The automated system is the only one of the three designed to meet the single-lane rate.

Cohen's kappa is calculated from observed agreement Po and chance agreement Pe: $\kappa = (Po - Pe)/(1 - Pe)$. For two human graders, construct a 2×2 table of accept/reject decisions on the same labelled battery set. For intra-rater agreement, compare the same operator's first and second round. For the automated system, repeat identical images and confirm deterministic consistency; compare accuracy separately against verified ground truth.

Agreement measure	**How to calculate**	**Interpretation**
Inter-rater κ	κ(G1,G2), κ(G1,G3), κ(G2,G3); report mean and range.	Measures agreement between different inspectors after correcting for chance.
Intra-rater κ	κ(G1 round 1, G1 round 2), repeated for each inspector.	Measures consistency of the same inspector over time.

Automated repeatability	Fraction of identical images receiving identical output on repeated runs.	Should approach 100% for deterministic processing; does not establish correctness.
Accuracy / recall	Correct decisions / all decisions; true defects detected / all true defects.	Recall is especially important because a missed unsafe lithium cell is a safety failure.

1.3 Input–output differences of related fields

Field	Input	Operation	Output in this project
Image Processing	Raw pixel array	Denoising, contrast correction, thresholding, morphology	A cleaner image or mask; no semantic decision is required.
Computer Vision	Image or video	Extracts shape, text, colour and defect meaning	Cell type, chemistry label, dimensions and safety status.
Machine Vision	Camera + controlled lighting + optics + industrial controller	Performs inspection and connects decisions to PLC / reject hardware	A timed reject signal to the pneumatic piston.
Computer Graphics	3-D models, geometry or scene description	Renders synthetic images or visual explanations	Simulated battery images for testing and system diagrams.

1.4 Decision

Technique C is selected for the production line because it supports non-contact operation, repeatable measurements, automatic traceability and the 15 cells/s rate. Technique B is useful as an operator review station and for exception handling. Manual inspection remains useful for audits and unusual defect review, but it should not control every production decision. A rule-based system must be monitored for new defect types and periodically validated with a labelled real-cell dataset.

TASK 2 – System Architecture: Pipeline Level Depth Comparison [5 Marks]

A low-level pixel rule can detect dark pixels, but it cannot reliably distinguish a true rust pit from a shadow, oil mark or touching battery. Mid-level geometry and a high-level classifier provide more context.

2.1 Three pipeline configurations

Technique	Processing levels	Operations	Strength / weakness
A – Low-level only	Low	Gaussian blur, adaptive threshold, count dark pixels outside a nominal region.	Very fast and simple; shadow edges and oil smudges can trigger false defects.
B – Low + mid-level	Low + mid	Thresholding, contours, minAreaRect, aspect ratio, moments, circularity, connected components.	Separates shapes and measures dimensions; fixed rules can fail on unusual poses or mixed defects.
C – Low + mid + high-level	Low + mid + high	Extract feature vector, then SVM or k-NN classifies cell type and safety class with confidence.	Best structured approach for known classes; needs representative labelled training examples and validation.

2.2 Level-annotated pipeline

LEVEL 0 – ACQUISITION

Global-shutter camera + diffuse LED strobe + encoder trigger

|

v

LEVEL 1 – LOW-LEVEL (pixels -> pixels)

BGR/RGB frame -> flat-field correction -> denoise -> HSV/Lab

-> adaptive threshold / edge map / binary foreground mask

|

v

LEVEL 2 – MID-LEVEL (pixels -> geometric/text features)

Contours -> connected components -> minAreaRect -> aspect ratio

-> circularity / moments / dimensions -> OCR crop and OCR text

|

v

LEVEL 3 – HIGH-LEVEL (features -> decision)

Feature vector + OCR chemistry + cell geometry

-> SVM / k-NN + explicit safety rules + confidence threshold

|

v

OUTPUT

Cell class + chemistry + defect list + confidence + timestamp

|

v

PLC / encoder tracking -> reject command -> pneumatic piston

2.3 Accuracy and latency comparison

The assignment does not include a labelled test set or benchmark execution output. The following table is therefore a planning estimate, not a measured result. It indicates the expected trend and the measurements that must be collected during commissioning.

Test case	**A: Low-level estimated accuracy**	**B: Low+mid estimated accuracy**	**C: Three-level target accuracy**
Separated clean cells	90–95%	96–98%	≥98.5%
Touching / side-by-side cells	55–70%	85–93%	≥97%
Oil smudges	60–75%	88–94%	≥97%
Strong shadows	50–70%	85–92%	≥97%
Overall design target	Not acceptable for safety release	Near target after tuning	Must be validated to ≤1.5% error

Pipeline	**Indicative CPU latency**	**At 15 cells/s?**	**Reason**

	per frame		
A – Low-level	2–5 ms	Yes	Few pixel operations; high false-trigger risk.
B – Low + mid	8–20 ms	Yes	Contours and geometric calculations add work.
C – Three-level classical ML	15–35 ms	Yes, if optimized	Feature extraction and classifier add latency; OCR may dominate.

The 15 cells/s arrival interval is 66.7 ms. The three-level pipeline should therefore be optimized to remain below 66.7 ms per cell, with a practical target of 35 ms or less to leave time for queueing, communication and reject tracking. Actual latency must be profiled on the specified four-core PC.

2.4 Why shadows cause false triggers

A low-level rule such as count(pixels where gray < T) assumes dark pixels are defects. A shadow also lowers intensity and creates a sharp boundary. If the threshold is fixed, a large shadow can produce more dark pixels than a small rust pit and exceed the defect count. Adaptive thresholding reduces slow illumination changes but can still classify a shadow edge as a crack. A robust pipeline uses diffuse strobed lighting, flat-field correction, local contrast, geometry masks and colour/texture features. The classifier should use several features rather than a single darkness count.

2.5 Classical ML versus CNN / ViT

An SVM or k-NN uses explicit features such as length, width, aspect ratio, circularity, mean Lab colour, OCR confidence and defect area. It is lightweight and explainable, which suits a CPU-only edge PC. A CNN learns local visual features from labelled images; a Vision Transformer (ViT) uses attention to model relationships across image patches. End-to-end models can learn complex patterns but generally require more labelled data, model validation and compute. A pretrained model may reduce training needs, but domain shift from factory lighting, battery suppliers and surface finishes remains a risk. For this line, a three-level classical pipeline is the practical baseline; a CNN can be evaluated later if a representative dataset and compute budget become available.

TASK 3 – Digital Representation: Color Space Selection for Oxide Rust Analysis [4 Marks]

Rust is brownish-red while healthy steel is metallic grey. Raw RGB values change strongly with illumination and specular reflections. HSV separates hue from brightness, while CIE L*a*b* provides a perceptually meaningful colour difference from a calibrated healthy-steel reference.

3.1 Comparison of colour spaces

Technique	Representation	Effect of illumination	Engineering assessment
A – RGB/BGR	Three channels: red, green and blue.	All channels scale with illumination; specular highlights may saturate channels.	Simple and fast, but thresholds are unstable between 1,500 and 45,000 lux.
B – HSV	Hue, saturation and value.	Hue is more stable than raw RGB for moderate intensity changes; low saturation and highlights make hue unreliable.	Useful for brown/red tint segmentation when combined with S and V masks.
C – CIE L*a*b*	L* lightness, a* green–red, b* blue–yellow.	L* captures brightness; a* and b* describe chromatic difference. ΔE from a healthy reference can separate rust from neutral steel.	Best calibrated choice for rust colour difference, with glare masking and fixed lighting.

3.2 Illumination scaling and cluster dispersion

For a first-order model, observed intensity I' = kI, where k is 1.0 (baseline), 0.4 (dim) or 1.8 (intense noon). In RGB, the three values all change with k; a fixed channel threshold therefore shifts the cluster. In HSV, ideal hue is largely unchanged by uniform scaling until clipping/noise or very low saturation occurs, while V changes directly. In Lab, a* and b* are more stable than L* under moderate neutral illumination changes, but the measured ΔE still changes when the illuminant spectrum or reflections change. Calibration and diffuse lighting remain necessary.

Lighting condition	Scale k	RGB cluster	HSV cluster	Lab cluster
Baseline	1.0	Reference RGB values	Reference H; S,V baseline	Reference L*,a*,b*

Dim	0.4	All channels become smaller; noise has greater relative effect	H mostly retained for saturated pixels; V drops	L* drops; a*,b* more stable under neutral light
Bright noon	1.8	Channels may clip at 255; colour separation is lost	V may clip; highlight pixels need masking	L* rises; chromatic axes remain more useful but spectral changes still matter

Qualitative cluster map: RGB clusters stretch radially as illumination changes; HSV clusters spread mainly along V; Lab clusters spread mainly along L* for a neutral light change. The most reliable production method is a*–b* / ΔE analysis after white-balance and flat-field calibration, with low-saturation/high-glare pixels excluded.

3.3 ΔE calculation

For a reference healthy-steel colour (L_0^*, a_0^*, b_0^*) and a measured pixel (L*, a*, b*), the simple CIE76 distance is $\Delta E^*ab = \sqrt{[(L^*-L_0^*)^2 + (a^*-a_0^*)^2 + (b^*-b_0^*)^2]}$. A pixel is a rust candidate when its colour distance and red/brown direction exceed calibrated thresholds. The threshold must be learned from labelled clean and rusted steel samples, not selected from a generic colour chart.

3.4 NumPy array architecture

A standard OpenCV colour image is stored as a three-dimensional NumPy array with shape (height, width, channels), dtype uint8. For example, a 1080 × 1920 BGR image has shape (1080, 1920, 3). image[y, x] selects one pixel; image[y, x, 0] is its blue channel, 1 is green and 2 is red. Rows represent image height and columns represent image width.

3.5 Integer overflow proof

```
import numpy as np
import cv2

a = np.array([250, 240, 10], dtype=np.uint8)
b = np.array([20, 30, 250], dtype=np.uint8)

print(a + b)              # uint8 wraparound: [14, 14, 4]
print(cv2.add(a, b))      # saturated: [255, 255, 255]
```

```
safe = a.astype(np.uint16) + b.astype(np.uint16)
print(safe)                 # [270, 270, 260]
```

An 8-bit unsigned integer can represent only 0 to 255. NumPy addition performed in uint8 wraps modulo 256: 250 + 20 = 270, and 270 mod 256 = 14. This can turn a bright pixel into a dark value and create false defects. cv2.add uses saturated arithmetic and clamps values above 255 to 255. For quantitative calculations, convert to uint16 or float32 before arithmetic; use cv2.add when saturation is the intended operation.

3.6 Decision

Technique C (Lab ΔE) is selected for calibrated rust detection because it measures distance from a known healthy alloy colour rather than relying on a single raw channel. HSV is a useful fast fallback and RGB is retained for display and compatibility. Diffuse lighting, white-balance references and highlight rejection are mandatory.

TASK 4 – Topological Analysis: Pixel Connectivity for Casing Scratch Counting [4 Marks]

The system must count distinct scratches and measure rust spots. Connectivity determines whether neighbouring foreground pixels belong to the same connected component.

4.1 Connectivity definitions

Technique	Neighbourhood	Behaviour	Risk
A – 4-connectivity (N4)	Up, down, left, right.	Diagonal pixels are separate components.	A one-pixel diagonal scratch is fragmented.
B – 8-connectivity (N8)	N4 plus four diagonal neighbours.	Diagonal lines remain connected.	Diagonal touching objects may merge at corner contacts.
C – m-connectivity	N4 plus selected diagonal links only when common 4-neighbours do not create ambiguity.	Reduces ambiguous diagonal loops in mixed patterns.	Requires custom neighbour testing and careful foreground/background convention.

4.2 Synthetic patterns and expected component counts

The following small patterns use 1 for foreground and 0 for background. Counts assume 8-connected background is not being counted; only foreground components are counted.

Pattern	Binary pattern	N4 components	N8 components	Interpretation
Diagonal line	100 / 010 / 001	3	1	N8 preserves a diagonal scratch.
Horizontal line	11111	1	1	Both connect straight scratches.
Two diagonal pixels	100 / 001	2	2	No adjacency exists.
Checkerboard 2×2	10 / 01	2	1	N8 joins diagonal foreground pixels.
Corner-	110 / 010 / 011	1	1	N4 path connects through

touching blobs				centre.

4.3 Python implementation: N4, N8 and m-connectivity

```
import numpy as np
from collections import deque

N4 = [(-1,0), (1,0), (0,-1), (0,1)]
DIAG = [(-1,-1), (-1,1), (1,-1), (1,1)]

def components(binary, mode="8"):
    a = (np.asarray(binary) > 0)
    h, w = a.shape
    seen = np.zeros_like(a, dtype=bool)
    labels = np.zeros((h, w), dtype=np.int32)
    label = 0
    for y in range(h):
        for x in range(w):
            if not a[y, x] or seen[y, x]:
                continue
            label += 1
            q = deque([(y, x)])
            seen[y, x] = True
            labels[y, x] = label
            while q:
                cy, cx = q.popleft()
                neighbours = list(N4)
                if mode == "8":
                    neighbours += DIAG
                elif mode == "m":
                    for dy, dx in DIAG:
                        ny, nx = cy + dy, cx + dx
                        if 0 <= ny < h and 0 <= nx < w and a[ny, nx]:
```

```
                    common = [(cy, nx), (ny, cx)]
                    if not any(0 <= yy < h and 0 <= xx < w and a[yy, xx]
                               for yy, xx in common):
                        neighbours.append((dy, dx))
            for dy, dx in neighbours:
                ny, nx = cy + dy, cx + dx
                if 0 <= ny < h and 0 <= nx < w and a[ny, nx] and not seen[ny, nx]:
                    seen[ny, nx] = True
                    labels[ny, nx] = label
                    q.append((ny, nx))
    return label, labels

p = np.array([[1,0,0],[0,1,0],[0,0,1]], dtype=np.uint8)
print(components(p, "4")[0])
print(components(p, "8")[0])
print(components(p, "m")[0])
```

4.4 Euclidean, City-Block and Chessboard distance

Metric	Formula	Meaning	Use
Euclidean De	$\sqrt{(dx^2 + dy^2)}$	Straight-line distance.	Best estimate of physical diameter and scratch width.
City-Block D4	$\|dx\| + \|dy\|$	Horizontal/vertical path length.	Overestimates diagonal distance; useful for N4 topology.
Chessboard D8	$\max(\|dx\|, \|dy\|)$	King-move distance.	Fast approximation for 8-neighbour geometry.

A 45° line with a true physical length L has horizontal and vertical projections $L/\sqrt{2}$. Euclidean distance reconstructs $\sqrt{[(L/\sqrt{2})^2+(L/\sqrt{2})^2]} = L$. D4 gives $\sqrt{2}$ L and overestimates by 41.4%; D8 gives $L/\sqrt{2}$ and underestimates by 29.3%. Therefore, Euclidean distance is the correct metric for calibrated physical dimensions.

4.5 OpenCV distance-transform code

```
import cv2
import numpy as np

binary = cv2.imread("battery_mask.png", cv2.IMREAD_GRAYSCALE)
_, mask = cv2.threshold(binary, 127, 255, cv2.THRESH_BINARY)

d_e = cv2.distanceTransform(mask, cv2.DIST_L2, 5)
d_l1 = cv2.distanceTransform(mask, cv2.DIST_L1, 3)
d_chess = cv2.distanceTransform(mask, cv2.DIST_C, 3)

mm_per_pixel = 0.078125
max_radius_mm = float(d_e.max()) * mm_per_pixel
equivalent_diameter_mm = 2 * max_radius_mm
print("Equivalent diameter (mm):", equivalent_diameter_mm)
```

4.6 Decision

Use 8-connectivity for thin diagonal scratches, then apply component area, length, width and shape filters to avoid merging unrelated corner contacts. Use m-connectivity when diagonal ambiguity is observed in checkerboard-like patterns. Use Euclidean distance for physical size because the belt calibration is in millimetres and must not depend on scratch orientation.

TASK 5 – Physical Acquisition: Sensor Shutter and Exposure Topology [5 Marks]

The cell moves at 1.2 m/s. Exposure time determines motion blur. A short exposure and synchronized strobe are needed to preserve the 0.3 mm defect and 0.4 mm text stroke.

5.1 Motion blur calculations

Motion blur distance b = v × t. With v = 1.2 m/s = 1,200 mm/s:

Exposure	Blur distance	Blur at FHD (0.1667 mm/px)	Blur at 12 MP (0.078125 mm/px)
4.0 ms	1,200 × 0.004 = 4.80 mm	28.8 px	61.44 px
1.0 ms	1,200 × 0.001 = 1.20 mm	7.2 px	15.36 px
0.05 ms (50 µs)	1,200 × 0.00005 = 0.060 mm	0.36 px	0.768 px

A 4 ms exposure smears a small rust pit across several millimetres and is unacceptable. A 1 ms exposure still causes 1.2 mm blur. A 50 µs exposure limits blur to 0.06 mm, below the 0.3 mm defect diameter. The recommended exposure is therefore approximately 50 µs or shorter, provided the LED strobe supplies sufficient light.

5.2 Shutter comparison

Technique	Advantages	Limitations	Decision
A – Rolling shutter + constant light	Low cost; common area-scan camera.	Rows are exposed at different times; moving cells skew. 100 Hz flicker can make rows differ in brightness.	Reject for precision inspection.
B – Global shutter + synchronized LED strobe	All pixels capture at the same instant; short pulse freezes motion; encoder synchronizes exposure.	Needs strobe driver, trigger wiring and sufficient peak LED power.	Best balanced choice.
C – Line-scan	Captures one line at a time;	Needs precise encoder	Useful for

camera	excellent for continuous web/conveyor imaging.	synchronization and line-rate configuration; more complex calibration and processing.	future high-speed scale-up or long continuous surfaces.

5.3 Rolling-shutter distortion and 100 Hz flicker

A 100 Hz light flicker has a period T = 1/100 = 10 ms. If the rolling-shutter sensor reads rows over a significant fraction of this period, different rows see different illumination phases. A moving cell also changes position between the first and last row exposure, creating a slanted or stretched shape. The exact distortion depends on the sensor's row-readout time, which is not specified; it must be measured from the selected camera datasheet and verified using a moving target.

5.4 Signal chain

Incident photons

|

v

Photodiode converts photons to charge

|

v

Charge integration during exposure

|

v

Charge-to-voltage conversion

|

v

Analog signal conditioning / correlated double sampling

|

v

ADC digitization (e.g., 8-bit or 12-bit)

|

v

Digital image buffer -> GigE / USB interface -> edge PC

5.5 CCD versus CMOS

CMOS is preferred for this industrial line because modern global-shutter CMOS sensors provide high frame rates, low power, compact integration and on-sensor readout circuitry. CCD can provide uniform image quality in some applications, but typically has higher power and readout constraints for this high-speed embedded system. The important selection criterion is not the historical sensor label alone: choose a global-shutter industrial sensor with low read noise, high quantum efficiency, reliable trigger input and a verified exposure/strobe mode.

5.6 Decision

Technique B is selected. Use a global-shutter CMOS area camera with encoder-triggered LED strobe at about 50 µs exposure. The strobe must be shielded from ambient light and the camera should be triggered at the required cell rate. Technique C remains a scale-up option where continuous line imaging is advantageous.

TASK 6 – Spatial Sampling: Resolution Optimization vs. Data Throughput [5 Marks]

The horizontal FOV is 320 mm. The pixel scale is calculated as FOV width divided by horizontal pixel count. The minimum defect is 0.3 mm and OCR stroke width is 0.4 mm.

6.1 Resolution calculations

Camera	mm/pixel	Samples across 0.3 mm defect	Samples across 0.4 mm stroke	RGB bandwidth at 15 fps (MB/s)	5-day RGB storage (TB)
VGA 640×480	0.5000	0.60	0.80	13.82	5.97
Full HD 1920×1080	0.1667	1.80	2.40	93.31	40.31
12 MP 4096×3000	0.0781	3.84	5.12	552.96	238.88

Calculations use one uncompressed 24-bit RGB image per cell at 15 fps for five continuous days (432,000 seconds). Storage = width × height × 3 bytes × 15 × 432,000. The table is a worst-case continuous archive assumption; event-triggered capture and lossless compression reduce the actual requirement.

6.2 Nyquist sampling

The Nyquist criterion requires at least two samples across the smallest feature to represent its spatial frequency. For a 0.4 mm text stroke, the theoretical minimum is 2 pixels, which requires mm/px ≤0.2 mm/px. FHD provides 2.4 pixels across the stroke and is just above the theoretical limit; 12 MP provides 5.12 pixels and is more robust for OCR. For a 0.3 mm defect, FHD provides only 1.8 pixels, below two samples; 12 MP provides 3.84 pixels. Thus 12 MP is the safer spatial choice for the stated defect size, but its raw bandwidth exceeds Gigabit Ethernet.

6.3 Frame rate requirement

A 70 mm cell passing through the 240 mm longitudinal FOV remains fully inside the view for (240−70)/1,200 = 0.1417 s. The geometric minimum to obtain at least one complete-frame opportunity during that interval is 1/0.1417 = 7.06 fps. However, the line throughput is 15 cells/s, so the camera and processing system must support at least 15 independent cell captures per second. Use 15 fps as the absolute throughput minimum and preferably 30 fps for timing margin.

6.4 Gigabit Ethernet and storage

Gigabit Ethernet has a theoretical line rate of 1,000 Mbit/s = 125 MB/s before protocol overhead. VGA and FHD RGB streams fit below this theoretical limit; 12 MP at 15 fps requires about 553 MB/s and cannot be sent uncompressed over one GigE link. A 12 MP system needs a faster interface, camera-side ROI, reduced frame rate, or compression. Five-day raw storage is far above the 1.5 TB SSD limit for all three resolutions.

6.5 Recommended archive policy

- Use 12 MP or a sensor with at least 4,096 horizontal pixels for detailed inspection, if the budget and interface allow.
- Store a full-resolution lossless image only for each detected defect, ambiguous OCR result and a controlled sample of accepted cells.
- Store a compressed audit image plus metadata for every cell; keep raw frames in a short rolling buffer.
- Retain measurements, OCR string, confidence, defect masks, timestamp and reject decision for every cell for the required five days.

6.6 Decision

VGA is inadequate for the 0.3 mm defect and 0.4 mm OCR stroke. FHD is a practical low-cost baseline but does not reliably sample the smallest defect. 12 MP provides the best spatial evidence but requires a faster interface and selective archival. A balanced system should use the highest resolution that can be sustained by the acquisition interface and storage policy.

TASK 7 – Brightness Quantization: Dynamic Range and Bruise/Dent Detection [4 Marks]

The smallest rust and dent contrast is only 5–8 grey levels on an 8-bit scale. Quantization must preserve these small differences while controlling the image data volume.

7.1 Quantization comparison

Bit depth	Number of levels	Approx. step if mapped from 0–255	Effect on 5–8 level contrast	Assessment
4-bit	16	17.0	Contrast may collapse into the same bin or one bin.	Not suitable for subtle defects.
6-bit	64	4.05	Only about 1–2 quantization steps remain.	Risky for low-contrast defects.
8-bit	256	1.0	5–8 distinct levels are preserved.	Minimum practical standard for this task.
12-bit	4096	0.0623	About 80–128 quantization steps represent 5–8 8-bit-equivalent levels.	Best headroom for calibration and later processing.

7.2 Quantization example

For uniform quantization, the step size is $\Delta = (I_{max} - I_{min})/(2^b - 1)$. If the full 0–255 range is mapped to 4 bits, $\Delta \approx 17$; a 5–8 level difference can disappear. At 6 bits, $\Delta \approx 4.05$; the defect may occupy only one or two levels and becomes sensitive to noise. At 8 bits, the 5–8 levels remain visible. A 12-bit sensor has 4096 codes and can preserve small variations, provided the analog noise and illumination stability are sufficiently low.

7.3 False contouring

When a smooth curved shoulder is reduced to 4 or 6 bits, many nearby intensity values are forced into the same output level. The smooth gradient becomes a series of flat bands. These artificial boundaries are called false contours or banding. An edge detector may interpret the bands as dents or pouch folds. Dithering can make banding less visible but cannot recover information that was discarded.

7.4 Image bit depth versus model quantization

Concept	Meaning	Example	Effect
Image quantization	Number of intensity values stored per pixel.	12-bit sensor data converted to 8-bit PNG.	Changes captured image precision and dynamic range.
Model quantization	Numeric precision used for neural-network weights/activations.	FP32 model converted to INT8 for edge deployment.	Reduces model memory and may speed inference; can slightly change model accuracy.

7.5 Storage trade-off

For the same pixel count, 12-bit packed data is 1.5 times the size of 8-bit data; if stored in 16-bit containers, it is twice the size. Under a strict 1.5 TB storage budget, preserve spatial detail for the 0.3 mm defect and use lossless compression, selective raw retention and metadata. Do not reduce to 4-bit or 6-bit because the 5–8 level defect contrast is too small. An 8-bit inspection stream is the practical baseline; 12-bit capture is preferable where the sensor and budget permit, followed by calibrated conversion for processing.

7.6 Decision

Choose 8-bit as the minimum production depth and 12-bit raw acquisition for high-risk inspection zones if the selected camera supports it. Avoid 4-bit and 6-bit capture. Preserve spatial samples and use a selective archive rather than sacrificing subtle intensity differences.

TASK 8 – Image Formation: Geometric Projection and Photometric Corrections [5 Marks]

The camera is mounted 400 mm above the belt and must cover a 320 mm horizontal field. Lens selection and calibration determine whether cell dimensions can be measured accurately.

8.1 Focal length calculation

A nominal 1/2-inch industrial sensor has an active width of approximately 6.4 mm (the exact active width must be confirmed from the selected sensor datasheet). Using similar triangles: FOV = sensor_width × distance / focal_length. Therefore f = 6.4 × 400 / 320 = 8.0 mm.

An approximately 8 mm lens is the first-order choice. The final lens must be checked for actual sensor dimensions, distortion, focus range and required image circle. A low-distortion lens and checkerboard calibration are required.

8.2 Perspective projection error

Perspective projection is x = fX/Z. The calibration is performed on the belt at Z_0=400 mm. A battery top surface is 30 mm above the belt, so its distance is Z=370 mm. The image magnification relative to the belt calibration is Z_0/Z = 400/370 = 1.08108. A true 30 mm feature on the top surface appears as 30×1.08108 = 32.43 mm when interpreted with the belt-plane scale. The error is +2.43 mm, or +8.11%.

This is too large for a ±0.5 mm pouch swelling requirement. A telecentric lens reduces magnification change with object height; alternatively, use a height-aware calibration or measure the pouch profile using a calibrated 3-D method.

8.3 Optics and lighting comparison

Technique	Configuration	Advantages	Limitations
A	Standard perspective lens + ambient light.	Lowest cost and simple installation.	Ambient varies from 1,500 to 45,000 lux; glare and shadows change thresholds; perspective error remains.
B	Standard lens + diffuse LED dome + checkerboard calibration.	Stable illumination, reduced specular glare, distortion correction and repeatable pixel scale.	Perspective magnification still changes with object height.
C	Object-space telecentric lens + diffuse dome.	Nearly constant magnification over a specified depth range; strongest dimensional accuracy.	High cost, limited field/depth range and possible budget conflict.

8.4 Flat-field normalization

A flat-field image F(x,y) records the illumination profile using a uniform reference. For an acquired image I(x,y), a corrected image can be computed as Icorr = I × mean(F) / (F + ε). The small ε prevents division by zero. The result reduces gradual brightness variation across the field.

```
import cv2
import numpy as np

img = cv2.imread("battery_frame.png").astype(np.float32)
flat = cv2.imread("uniform_reference.png").astype(np.float32)

mean_flat = np.mean(flat, axis=(0, 1), keepdims=True)
corrected = img * mean_flat / np.maximum(flat, 1.0)
corrected = np.clip(corrected, 0, 255).astype(np.uint8)

cv2.imwrite("flat_field_corrected.png", corrected)
```

8.5 Homomorphic filtering

The image formation model can be written as f(x,y)=i(x,y)r(x,y), where i is illumination and r is reflectance. Taking logarithms gives log f = log i + log r. Homomorphic filtering suppresses slowly varying illumination (low-frequency component) while preserving reflectance detail (higher-frequency component). It is useful for uneven lighting, but it does not replace a diffuse dome or a controlled strobe.

8.6 Decision

Technique B is the balanced production choice: 8 mm calibrated lens, diffuse LED dome and checkerboard distortion calibration. If the ±0.5 mm pouch measurement must be guaranteed across a 30 mm height range, Technique C or an additional height-aware measurement is required. The cost of a telecentric lens must be checked against the ₹1,25,000 cap.

TASK 9 – Systems Integration: Balanced Front-End Optimization Challenge [6 Marks]

The final system must combine sensing, lighting, optics, processing, storage and the reject actuator. The rejector is 0.48 m downstream, giving exactly 400 ms at 1.2 m/s.

9.1 Three system packages

Component	A – Budget setup	B – Balanced setup	C – Premium setup
Camera	1.3 MP rolling shutter USB	2 MP global shutter GigE	4k high-speed line-scan
Lighting	Ambient factory light	Diffuse LED dome + synchronized strobe	Uniform high-power line light
Lens	Standard lens	Calibrated standard low-distortion lens	High-precision telecentric lens
Archive	8-bit lossy JPEG	8-bit lossless PNG + event metadata	12-bit raw
Software	Low + mid-level	Low + mid + classical ML/OCR	Three-level pipeline
Main risk	Blur, flicker, glare and compression artefacts.	2 MP sampling may be marginal for 0.3 mm defect.	High cost, data rate and CPU/storage pressure.

9.2 Indicative budget allocation

The following is a planning allocation, not a vendor quotation. Industrial prices vary by sensor, interface, lens, lighting power, enclosure, trigger accessories, taxes and support. Obtain written quotations before procurement.

Item	Balanced design allocation (₹)	Reason
2 MP global-shutter GigE camera	35,000	Industrial trigger support and suitable frame rate.
8 mm low-distortion lens	8,000	320 mm FOV at 400 mm standoff, subject to sensor width.
Diffuse LED dome / ring lighting	18,000	Reduces specular glare and ambient

		variation.
Strobe controller + encoder trigger	12,000	Short synchronized exposure.
Mount, enclosure, cables and I/O	12,000	Stable alignment and industrial protection.
Storage / SSD and backup allocation	8,000	Audit image and metadata retention.
Integration / contingency	20,000	PLC interface, calibration and spare allowance.
TOTAL	1,13,000	Within ₹1,25,000 cap; ₹12,000 reserve.

The allocation leaves ₹12,000 contingency. The selected PC is assumed to be the already available four-core system and is not charged to the imaging hardware budget. Actual quotes may exceed this estimate; the design must be re-costed before purchase.

9.3 End-to-end pipeline and time budget

Photons
-> Global-shutter sensor + 50 us LED strobe
-> ADC / camera frame buffer
-> GigE transfer to edge PC
-> Flat-field correction + denoise + colour conversion
-> Segmentation + contours + dimensions
-> OCR chemistry code + SVM / safety rules
-> Cell ID + confidence + defect decision
-> Encoder-based tracking queue
-> PLC output
-> Pneumatic reject piston

Stage	**Allocated time (ms)**	**Cumulative time (ms)**
Exposure / strobe capture	0.05	0.05
Sensor readout / frame availability	8	8.05

GigE transfer / buffering	10	18.05
Pre-processing and segmentation	12	30.05
Geometry and defect measurements	18	48.05
OCR and chemistry classification	35	83.05
High-level decision / confidence checks	8	91.05
Queue / encoder tracking and PLC communication	10	101.05
Actuator command margin	20	121.05
TOTAL allocated	121.05	121.05

The planned decision path is about 121 ms, leaving approximately 279 ms before the 400 ms deadline. These are engineering allocations, not measured timings. The system must be profiled on the actual four-core PC under worst-case OCR and image-load conditions. The arrival interval is 66.7 ms, so the pipeline must also be pipelined and sustain at least 15 cells/s; deadline compliance alone does not prove throughput.

9.4 Engineering decision matrix

Condition	A – Budget	B – Balanced	C – Premium
Passing storm clouds	Ambient illumination changes thresholds; high risk.	Enclosed dome and strobe isolate the image from ambient changes.	Line light and enclosure offer strong control.
Vibration	Rolling-shutter skew and motion blur worsen.	Global shutter captures all rows simultaneously; rigid mount required.	Line-scan encoder synchronization is sensitive to vibration.
Specular glare	Uncontrolled bright highlights.	Diffuse dome spreads reflections; cross-polarization may help.	Uniform line illumination can be tuned; telecentric optics aid geometry.
CPU / data load	Low, but accuracy is weak.	Moderate and manageable with ROI and classical ML.	High data volume; likely difficult on CPU-only PC.
Budget	Low initial cost.	Designed to remain under	Likely exceeds cap.

		cap with indicative allocation.	
Overall	Reject	Recommended balanced option	Technically strong but over-budget / over-complex.

9.5 Final selection

Technique B is selected as the best balance of image quality, deterministic timing, lighting stability, traceability and cost. The 2 MP resolution is a compromise and may not reliably resolve a 0.3 mm defect across the full 320 mm FOV. If the 1.5% miss-rate requirement is strict, the site should either increase horizontal resolution, narrow the inspection FOV, or use multiple views while retaining the global-shutter/strobe architecture. The budget and sampling requirement must be reconciled before acceptance testing.

9.6 Applications transfer

Application	Reusable part of the design	Additional requirement
Food sorting	Global shutter, controlled illumination, colour and shape features.	Food-safe enclosure and product-specific colour calibration.
PCB inspection	High-resolution imaging, OCR, defect masks and traceability.	Macro optics, smaller FOV and component-level datasets.
Pharmaceutical packaging	OCR, code verification and reject tracking.	Validated serialization and regulatory audit controls.
Metal surface inspection	Lab colour, texture, flat-field and directional lighting.	Polarizers and surface-specific defect references.

TASK 10 – Adaptive Engineering: High-Speed Plant Scale-Up Challenge [8 Marks]

The new contract doubles conveyor speed to 2.4 m/s, increases throughput to 30 cells/s, halves the decision window to 200 ms and requires MWIR thermal detection of 0.5°C localized hot spots. The edge PC remains four-core CPU-only.

10.1 Recalculated timing and spatial constraints

At 2.4 m/s, the belt moves 2,400 mm/s. The rejector distance remains 0.48 m, so travel time is 0.48/2.4 = 0.200 s = 200 ms. The cell arrival interval is 1/30 = 33.3 ms. Processing must therefore sustain 30 cells/s and deliver each decision within 200 ms.

Parameter	Original line	Scaled line	Change
Belt speed	1.2 m/s	2.4 m/s	2× faster
Throughput	15 cells/s	30 cells/s	2× higher
Reject window	400 ms	200 ms	Half
Arrival interval	66.7 ms	33.3 ms	Half
FOV	320 × 240 mm	320 × 240 mm (assumed unchanged)	Same
12 MP pixel scale	0.078125 mm/px	0.078125 mm/px	Unchanged if resolution/FOV unchanged

10.2 Maximum exposure to prevent blur

To keep motion blur at or below 0.3 mm: tmax = 0.3 mm / 2,400 mm/s = 0.000125 s = 125 µs. To keep blur at or below one pixel at 12 MP (0.078125 mm/px): tmax = 0.078125/2,400 = 32.55 µs. A 30 µs synchronized strobe is therefore a suitable target for approximately one-pixel motion blur at 12 MP.

Exposure	Blur at 2.4 m/s	At 12 MP (0.078125 mm/px)
1.0 ms	2.40 mm	30.72 px
0.125 ms	0.30 mm	3.84 px
0.05 ms	0.12 mm	1.54 px
0.03 ms	0.072 mm	0.92 px

10.3 Data generation rate

At 30 fps, uncompressed RGB data rates double from Task 6. VGA: 640×480×3×30 = 27.65 MB/s. FHD: 1920×1080×3×30 = 186.62 MB/s. 12 MP: 4096×3000×3×30 = 1,105.92 MB/s. The 12 MP stream cannot fit on Gigabit Ethernet; even FHD exceeds the theoretical 125 MB/s GigE limit as uncompressed RGB. Use a faster camera interface, monochrome/ROI stream, compression or on-camera processing.

10.4 Strategy comparison

Strategy	Action	Advantages	Limitations
A – RETAIN + ROI	Keep current hardware; shorten exposure, crop ROIs and reduce boundary processing.	Lowest cost and fastest deployment; may improve CPU time.	Cannot detect invisible internal thermal defects with visible light. Existing camera may not support required exposure, frame rate or trigger performance.
B – MODIFY sensing	Upgrade to high-frame-rate global-shutter camera, ultra-short strobe, light tunnel and INT8 arrays where applicable.	Improves visible-light capture, reduces ambient interference and lowers compute load.	Still cannot see internal thermal hot spots using visible wavelengths; INT8 does not create thermal information.
C – REDESIGN architecture	Dual stream: high-speed visible line-scan plus synchronized MWIR thermal sensor in regulated enclosure.	Adds the required spectral information and supports high-speed visible inspection.	Highest integration cost; thermal calibration, synchronization and CPU inference must be engineered carefully.

10.5 MWIR thermal detection

Visible-light cameras detect reflected visible wavelengths. Internal short-circuit heating is specified as a subtle 0.5°C thermal variation and is not visible in the RGB image. A mid-wave infrared sensor detects emitted thermal radiation in its spectral band. The thermal sensor must have suitable NETD (noise-equivalent temperature difference), calibration stability, frame rate and spatial resolution to resolve a

0.5°C change under the enclosure's operating conditions. A regulated enclosure reduces ambient thermal drift and contamination.

10.6 Few-shot learning with a pretrained Vision Foundation Model

A pretrained Vision Foundation Model can provide general visual features. Few-shot transfer uses a small number of labelled examples of new thermal defect shapes to adapt a classifier or prompt-based detector. This can reduce the number of new training images compared with training a CNN from scratch. However, visible-light pretraining may not transfer directly to MWIR thermal textures. Domain shift can arise from sensor response, emissivity, battery casing, temperature range and factory background.

- Create a labelled MWIR validation set from safe, controlled test cells and known thermal references.
- Keep a frozen baseline and compare few-shot adaptation against a simple calibrated thermal threshold.
- Measure false negatives, false positives, inference latency and calibration drift separately.
- Do not let a model silently replace a safety rule; low-confidence or out-of-distribution cells should be diverted for safe review.

10.7 Edge execution and operational risks

Risk	Effect	Mitigation
Domain shift	New battery supplier or thermal background changes model features.	Periodic labelled validation; reference targets; drift alarms.
Traceability failure	Thermal image and visible cell ID may be mismatched.	Hardware timestamps, encoder position, synchronized triggers and immutable event logs.
Inference latency	Foundation model may exceed 200 ms on CPU.	Use compact feature extractor, quantized model, ROI and benchmark worst-case latency.
Thermal calibration drift	A 0.5°C signal may be confused with sensor drift.	Regulated enclosure, blackbody/reference check and scheduled calibration.
Data overload	30 cells/s dual streams generate	ROI, lossless event images,

	large data volume.	compressed audit evidence and retention policy.

10.8 RETAIN / MODIFY / REDESIGN decision

The recommended pivot is REDESIGN (Strategy C). Strategy A can improve speed but cannot satisfy the new MWIR thermal requirement. Strategy B improves visible sensing and timing but still cannot observe the internal thermal signal. Strategy C is the only strategy that adds the required MWIR stream while retaining visible inspection for shape, OCR and surface defects. The redesign must be staged: first validate thermal sensitivity and synchronization, then benchmark the dual-stream pipeline on the four-core PC. If a full foundation model is too slow, use a compact calibrated thermal detector for real-time rejection and reserve few-shot model analysis for offline review.

10.9 Final conclusion

The original balanced global-shutter/strobe architecture is appropriate for the 1.2 m/s line, but the scale-up changes both timing and sensing requirements. At 2.4 m/s, exposure should be about 30 µs to limit 12 MP motion blur to approximately one pixel, and the system must accept a new cell every 33.3 ms. The thermal short-circuit requirement cannot be solved by visible-light optimization or INT8 conversion. A synchronized visible + MWIR architecture is therefore required, with strict thermal calibration, event traceability and CPU latency testing.

OVERALL ENGINEERING CONCLUSION

For the original production line, the recommended front-end is a global-shutter CMOS area-scan camera with a short encoder-synchronized LED strobe, diffuse dome illumination, an approximately 8 mm low-distortion lens, checkerboard calibration and a three-level OpenCV/classical-ML pipeline. Lab ΔE is preferred for calibrated rust colour analysis; contours and Euclidean measurements are used for shape and defect dimensions; OCR is used for chemistry codes. The system should process each cell in less than the 66.7 ms arrival interval while issuing the reject decision before the 400 ms gate deadline.

The main engineering conflict is spatial resolution versus interface and storage. FHD is convenient but marginal for a 0.3 mm defect; 12 MP is more suitable spatially but exceeds GigE raw bandwidth and the five-day 1.5 TB archive. Event-based lossless evidence, compressed audit images and metadata are required. For the scale-up, the 200 ms decision window and 0.5°C thermal requirement force a dual-stream visible/MWIR redesign.

LIMITATIONS AND VALIDATION PLAN

- The assignment does not provide real battery images or ground-truth defect labels. All accuracy percentages and latency allocations in this answer are design targets/estimates, not measured outcomes.
- Camera sensor active dimensions, rolling-shutter row time, lens distortion, strobe intensity, OCR engine speed and PLC response must be verified from actual hardware.
- The ₹1,13,000 balanced package is an indicative allocation and must be replaced with supplier quotations before purchase.
- Acceptance testing must use representative 18650, 21700 and pouch cells, all four chemistry labels, clean and defective samples, multiple suppliers and the full lighting range.
- To verify the ≤1.5% miss/misclassification limit, report a confusion matrix with confidence intervals and separately measure false-negative rate for each hazardous defect category.

REFERENCES

- 1. R. C. Gonzalez and R. E. Woods, Digital Image Processing, 4th ed., Pearson, 2018.
- 2. D. A. Forsyth and J. Ponce, Computer Vision: A Modern Approach, 2nd ed., Pearson, 2011.
- 3. OpenCV Documentation, Image Processing, Geometric Image Transformations, Contours and Shape Analysis, https://docs.opencv.org/ (accessed 31 October 2026).

- 4. NumPy Documentation, Data types and array operations, https://numpy.org/doc/ (accessed 31 October 2026).
- 5. CIE, Colorimetry, 4th edition, CIE 015:2018.
- 6. EMVA, EMVA 1288 Standard for Characterization of Image Sensors and Cameras, European Machine Vision Association.
- 7. ISO 12233, Photography – Electronic still picture imaging – Resolution and spatial frequency responses.
- 8. Szeliski, R., Computer Vision: Algorithms and Applications, 2nd ed., Springer, 2022.
- 9. scikit-learn Documentation, Support Vector Machines and Nearest Neighbors, https://scikit-learn.org/stable/ (accessed 31 October 2026).
- 10. NIST, Guide to Industrial Control Systems (ICS) Security, SP 800-82 Rev. 3, 2023.

APPENDIX A – CORE IMAGE PROCESSING CODE

The following compact reference program demonstrates the low- and mid-level operations used by the proposed system. Production deployment requires camera-specific calibration, OCR integration, dataset validation and PLC safety interlocks.

```
import cv2
import numpy as np

def inspect_battery(frame, mm_per_pixel):
    blurred = cv2.GaussianBlur(frame, (5, 5), 0)
    lab = cv2.cvtColor(blurred, cv2.COLOR_BGR2LAB)
    hsv = cv2.cvtColor(blurred, cv2.COLOR_BGR2HSV)

    gray = cv2.cvtColor(blurred, cv2.COLOR_BGR2GRAY)
    mask = cv2.adaptiveThreshold(
        gray, 255, cv2.ADAPTIVE_THRESH_GAUSSIAN_C,
        cv2.THRESH_BINARY_INV, 31, 7
    )

    kernel = np.ones((3, 3), np.uint8)
    mask = cv2.morphologyEx(mask, cv2.MORPH_OPEN, kernel)
```

```
    mask = cv2.morphologyEx(mask, cv2.MORPH_CLOSE, kernel)

    contours, _ = cv2.findContours(
        mask, cv2.RETR_EXTERNAL, cv2.CHAIN_APPROX_SIMPLE
    )

    objects = []
    for c in contours:
        area = cv2.contourArea(c)
        if area < 30:
            continue
        rect = cv2.minAreaRect(c)
        w, h = rect[1]
        length_px = max(w, h)
        width_px = min(w, h)
        length_mm = length_px * mm_per_pixel
        width_mm = width_px * mm_per_pixel
        aspect = length_px / max(width_px, 1.0)
        perimeter = cv2.arcLength(c, True)
        circularity = 4 * np.pi * area / max(perimeter**2, 1.0)

        objects.append({
            "area": area,
            "length_mm": length_mm,
            "width_mm": width_mm,
            "aspect_ratio": aspect,
            "circularity": circularity
        })

    return objects, mask, lab, hsv

if __name__ == "__main__":
    image = cv2.imread("battery_frame.png")
```

```
if image is None:
    raise SystemExit("Image not found")
result, binary, lab, hsv = inspect_battery(image, 0.078125)
print(result)
cv2.imwrite("battery_mask.png", binary)
```

APPENDIX B – ACCEPTANCE TEST CHECKLIST

Test	Method	Pass condition
Cell form factor	Test 18650, 21700 and pouch cells at multiple orientations.	Correct class and dimensions within specified tolerance.
Chemistry OCR	Capture LCO, LFP, NMC and NiMH labels under glare and lighting variation.	Correct string or low-confidence reject; no silent wrong chemistry.
Rust / micro-crack	Use calibrated 0.3 mm reference defects and low-contrast rust samples.	Required defect recall demonstrated on held-out samples.
Pouch swelling	Use reference profiles / gauge measurements.	Dimensional error within ±0.5 mm.
Motion blur	Move a calibrated target at 1.2 m/s and verify exposure.	Blur remains within selected pixel/defect limit.
Flicker immunity	Test at 1,500–45,000 lux and 100 Hz lighting flicker.	No row banding or unacceptable threshold drift.
Throughput	Run continuous 15 cells/s stream.	No dropped cells; sustained processing below 66.7 ms per arrival.
Reject timing	Track cell to piston 0.48 m downstream.	Decision and actuator command within 400 ms.
Storage	Run five-day retention simulation.	Every cell has retrievable audit evidence within 1.5 TB.
Safety accuracy	Blind test with verified ground truth.	Misclassification / defect miss rate ≤1.5% with reported

		confidence.

DETAILED TECHNICAL EXPANSION – EXTENDED SUBMISSION

This section expands the original task answers with additional engineering explanation, design implications, validation procedures and implementation considerations. It follows the same ten-task structure as the assignment brief.

TASK 1 – EXTENDED TECHNICAL DISCUSSION

1.1 Operational requirements and risk classification

The sorting line has three separate decisions: identify the physical form factor, identify the chemistry marking, and determine whether the cell is safe to continue. These decisions must not be collapsed into one label. A cylindrical 18650 cell and a 21700 cell may share a similar appearance, but their dimensions affect handling guides. A chemistry error can contaminate a smelter batch, while a defect error can allow a damaged cell to continue through the process.

The inspection record should therefore contain a unique cell ID, image timestamp, form-factor result, chemistry result, defect result, confidence values and final action. A low-confidence chemistry result should lead to quarantine or rejection. This is safer than forcing a prediction when the visual evidence is incomplete.

Engineering implications

The production design should separate classification, safety assessment and actuator control. A decision must be traceable to the image and encoder position that produced it.

Practical validation checklist

- Test all three form factors and all chemistry labels.
- Include damaged, unreadable and uncertain cells.
- Measure per-class errors and confirm correct piston tracking.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

1.2 Manual inspection: strengths and failure mechanisms

Manual inspection has the advantage of human flexibility. A trained operator can recognize unusual shapes, interpret context and notice a condition that was not anticipated in the software. It also has low initial equipment cost. However, the assignment describes changing ambient illumination, small laser-etched characters and rapid eye fatigue. These factors directly reduce the reliability of manual inspection.

Human performance is not constant over a long shift. Attention, experience, viewing angle and fatigue can change the result. A second operator may classify the same borderline cell differently. Manual inspection is therefore appropriate for exception review, sample auditing and maintenance checks, but not as the sole inspection method at a sustained rate of 15 cells per second.

Engineering implications

The production design should separate classification, safety assessment and actuator control. A decision must be traceable to the image and encoder position that produced it.

Practical validation checklist

- Test all three form factors and all chemistry labels.
- Include damaged, unreadable and uncertain cells.
- Measure per-class errors and confirm correct piston tracking.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

1.3 Image-assisted inspection and human factors

Image-processing-assisted inspection places a camera between the object and the operator. Contrast enhancement, magnification and display overlays can make small chemistry codes easier to read. The camera can also save images for later review. This improves the information available to the operator but does not remove the human decision bottleneck.

The display system must present one cell at a time or provide an orderly queue with cell identifiers. If the screen shows an unstructured stream, the operator may lose track of which image belongs to which physical cell. A useful interface should highlight uncertain regions, show the original image beside the enhanced image, and record the operator's final decision.

Engineering implications

The production design should separate classification, safety assessment and actuator control. A decision must be traceable to the image and encoder position that produced it.

Practical validation checklist

- Test all three form factors and all chemistry labels.
- Include damaged, unreadable and uncertain cells.
- Measure per-class errors and confirm correct piston tracking.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

1.4 Automated machine vision and fail-safe control

An automated system connects the camera, illumination, processing software, object tracker and pneumatic reject mechanism. The computer applies the same algorithm to every cell and can maintain a digital audit trail. The key benefit is repeatability at the required line rate. Automation also reduces the need for direct human contact with potentially damaged cells.

Automation must be fail-safe. If a frame is missing, a cell cannot be tracked, or the chemistry classifier returns low confidence, the system must not silently pass the cell. A reject or quarantine action is safer. The controller should use encoder position and cell ID to ensure that a decision is applied to the correct battery rather than relying only on a fixed software delay.

Engineering implications

The production design should separate classification, safety assessment and actuator control. A decision must be traceable to the image and encoder position that produced it.

Practical validation checklist

- Test all three form factors and all chemistry labels.
- Include damaged, unreadable and uncertain cells.
- Measure per-class errors and confirm correct piston tracking.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

1.5 Repeatability, objectivity and agreement measurement

Repeatability asks whether the same inspector or system produces the same result when the same sample is evaluated repeatedly. Objectivity asks whether the result depends on who performs the inspection. Cohen's Kappa is a statistic for agreement between two raters that adjusts for agreement expected by chance. It is calculated as kappa = (Po - Pe)/(1 - Pe), where Po is observed agreement and Pe is chance agreement.

The assignment does not provide a labelled inspection dataset or two-rater results. Therefore, a numerical Kappa value must not be invented. The correct procedure is to select a representative set of cells, have two trained inspectors label them independently, calculate the confusion table, and compare the automated system with the adjudicated reference labels.

Engineering implications

The production design should separate classification, safety assessment and actuator control. A decision must be traceable to the image and encoder position that produced it.

Practical validation checklist

- Test all three form factors and all chemistry labels.
- Include damaged, unreadable and uncertain cells.
- Measure per-class errors and confirm correct piston tracking.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

1.6 Throughput, endurance and traceability

At 15 cells per second, the mean arrival interval is 1/15 second, or about 66.7 milliseconds. A single cell has 400 milliseconds to travel from the camera to the piston because the piston is 0.48 metres downstream and the belt moves at 1.2 metres per second. The 400 ms window is not the same as the arrival interval: the software may pipeline several cells at once, but it must retain each cell's identity.

Traceability requires the system to retain enough information to reconstruct a decision. Useful fields include camera ID, encoder count, image timestamp, software version, calibration version, class scores, defect measurements, operator override and actuator output. The record should make it possible to identify whether a failure came from acquisition, classification, tracking or actuation.

Engineering implications

The production design should separate classification, safety assessment and actuator control. A decision must be traceable to the image and encoder position that produced it.

Practical validation checklist

- Test all three form factors and all chemistry labels.
- Include damaged, unreadable and uncertain cells.
- Measure per-class errors and confirm correct piston tracking.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

1.7 Selection and validation plan

Fully automated machine vision is selected as the primary workflow because it is the only option designed to sustain 15 cells per second with consistent decisions and automatic evidence capture. Image-assisted inspection remains useful as a review station for low-confidence cases. Manual inspection remains useful for maintenance and independent sample checks.

Before release, the system should be tested with clean cells, damaged cells, unreadable codes, touching objects, shadows, oil marks and reflective surfaces. Report per-class precision and recall, defect miss rate, false-reject rate, dropped-frame count and 95th-percentile latency. The assignment's maximum miss-rate requirement of 1.5% must be demonstrated using representative test data.

Engineering implications

The production design should separate classification, safety assessment and actuator control. A decision must be traceable to the image and encoder position that produced it.

Practical validation checklist

- Test all three form factors and all chemistry labels.
- Include damaged, unreadable and uncertain cells.
- Measure per-class errors and confirm correct piston tracking.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

TASK 2 – EXTENDED TECHNICAL DISCUSSION

2.1 Low-level processing and pixel operations

Low-level image processing works directly with pixel values. Typical operations include colour conversion, smoothing, thresholding, edge detection and morphology. Gaussian blur reduces high-frequency noise before thresholding. Adaptive thresholding calculates a local threshold so that the same global intensity cut-off is not applied across an image with uneven brightness.

A low-level dark-pixel counter is simple and fast, but it has no semantic understanding. A shadow, oil stain, printed marking or reflection can produce dark pixels even when the battery surface is intact. Conversely, a real rust spot may be missed if its intensity is close to the surrounding casing. Pixel counts should therefore be treated as candidate evidence rather than a final safety decision.

Engineering implications

Use a staged pipeline so that expensive classification is performed only after image quality and object segmentation have been checked. Keep intermediate masks and feature measurements for debugging.

Practical validation checklist

- Run the same labelled images through each pipeline.
- Include shadows, oil, touching objects and glare.
- Record accuracy, miss rate and 95th-percentile latency.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

2.2 Adaptive thresholding and parameter selection

Adaptive thresholding calculates a local threshold from a neighbourhood around each pixel. The block size controls the size of that neighbourhood, while the constant C shifts the threshold. A small block size reacts to fine local changes but may amplify texture and noise. A large block size responds to broader illumination changes but can remove small defects or merge nearby regions.

Parameter selection should be performed using a labelled set containing clean metallic surfaces, rust, oil, shadow and etched text. The same parameters should then be tested on images captured at the minimum and maximum ambient light conditions. Parameters should not be tuned only on a single attractive example image.

Engineering implications

Use a staged pipeline so that expensive classification is performed only after image quality and object segmentation have been checked. Keep intermediate masks and feature measurements for debugging.

Practical validation checklist

- Run the same labelled images through each pipeline.
- Include shadows, oil, touching objects and glare.
- Record accuracy, miss rate and 95th-percentile latency.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

2.3 Mid-level geometry and contour features

Mid-level processing groups pixels into regions and describes their geometry. Contours provide boundaries; contour area measures the enclosed pixel area; a minimum-area rectangle estimates width, height and orientation; and moments provide centroid and shape information. Circularity can be calculated as $4\pi A/P^2$, where A is area and P is perimeter.

These measurements help distinguish long cylindrical projections from broad pouch-like objects. They also provide defect descriptors such as component length, width, area and orientation. The measurements are sensitive to segmentation quality: a merged shadow can enlarge a contour, while a broken diagonal scratch can be measured as several small components.

Engineering implications

Use a staged pipeline so that expensive classification is performed only after image quality and object segmentation have been checked. Keep intermediate masks and feature measurements for debugging.

Practical validation checklist

- Run the same labelled images through each pipeline.
- Include shadows, oil, touching objects and glare.
- Record accuracy, miss rate and 95th-percentile latency.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

2.4 Touching batteries and segmentation ambiguity

When two batteries touch, a binary segmentation mask may contain one connected region instead of two. A single contour then gives incorrect area and aspect ratio. Possible remedies include watershed separation, distance-transform peaks, convexity analysis, controlled mechanical spacing and object tracking across successive frames.

Mechanical separation is often more reliable than trying to solve every touching-object case in software. If the process cannot guarantee spacing, the algorithm should detect an ambiguous merged region and route it to a safe reject path. It should not confidently assign a single-cell class to a multi-cell contour.

Engineering implications

Use a staged pipeline so that expensive classification is performed only after image quality and object segmentation have been checked. Keep intermediate masks and feature measurements for debugging.

Practical validation checklist

- Run the same labelled images through each pipeline.
- Include shadows, oil, touching objects and glare.
- Record accuracy, miss rate and 95th-percentile latency.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

2.5 High-level classification using SVM or k-NN

A classical classifier receives a feature vector rather than raw pixels. The vector may include aspect ratio, area, circularity, colour statistics, edge density, OCR confidence and measured dimensions. An SVM learns a separating boundary between classes. k-NN assigns a label based on the nearest training examples in feature space.

Features should be scaled before SVM training because area and circularity can have very different numerical ranges. Training data should include the full range of orientations, surface conditions and illumination conditions expected in production. Probability estimates must be calibrated and low-confidence predictions must trigger a safe action.

Engineering implications

Use a staged pipeline so that expensive classification is performed only after image quality and object segmentation have been checked. Keep intermediate masks and feature measurements for debugging.

Practical validation checklist

- Run the same labelled images through each pipeline.
- Include shadows, oil, touching objects and glare.
- Record accuracy, miss rate and 95th-percentile latency.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

2.6 Classical learning compared with CNN and ViT

A CNN learns spatial filters from labelled images and builds progressively more complex representations. A Vision Transformer divides an image into patches and uses attention to model relationships between patches. These methods can learn complex appearance patterns, but they usually require more training data and more compute than a compact feature-based classifier.

The assignment limits the computer to four CPU cores and no GPU. A feature-based SVM or k-NN model is a reasonable starting point because its inference cost is predictable and its features can be inspected. A compact neural model may be considered later only after benchmarking the complete pipeline on the target computer.

Engineering implications

Use a staged pipeline so that expensive classification is performed only after image quality and object segmentation have been checked. Keep intermediate masks and feature measurements for debugging.

Practical validation checklist

- Run the same labelled images through each pipeline.
- Include shadows, oil, touching objects and glare.
- Record accuracy, miss rate and 95th-percentile latency.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

2.7 Benchmark design and pipeline selection

The brief asks for accuracy and latency on difficult examples but does not provide a labelled test set. A valid comparison should run all three approaches on exactly the same held-out images. The dataset should include side-by-side cells, oil smudges, strong shadows, rust, dents, swelling and partly hidden chemistry markings.

Record confusion matrices, per-class recall, defect miss rate, false-reject rate, average latency, 95th-percentile latency and dropped-frame count. The three-level pipeline is selected because it combines simple pixel preparation, interpretable geometry and a learned decision boundary. Its compliance with the 1.5% miss-rate limit remains a test requirement, not an assumed result.

Engineering implications

Use a staged pipeline so that expensive classification is performed only after image quality and object segmentation have been checked. Keep intermediate masks and feature measurements for debugging.

Practical validation checklist

- Run the same labelled images through each pipeline.
- Include shadows, oil, touching objects and glare.
- Record accuracy, miss rate and 95th-percentile latency.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

3.1 Colour representation and illumination dependence

Colour values are affected by illumination intensity, spectrum, surface reflectance, camera response and exposure. A polished metal casing may reflect the light source rather than display its own material colour. Therefore, a fixed threshold in raw BGR channels can shift when the factory moves from cloudy light to intense afternoon light.

The assignment gives three relative illumination conditions: 0.4×, 1.0× and 1.8×. The same healthy and rusted samples should be imaged at all three levels. A robust representation should keep healthy and rust classes separated after controlled white balance and illumination correction.

Engineering implications

Lock acquisition settings and use a reference sample to detect colour drift. Colour thresholds must be calibrated with the actual casing material and production lighting.

Practical validation checklist

- Capture reference samples at 0.4×, 1.0× and 1.8× illumination.
- Compare class-centroid spread in each colour space.
- Check arithmetic saturation and colour calibration drift.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

3.2 Raw RGB/BGR channel analysis

RGB stores red, green and blue components. OpenCV conventionally reads a colour image in BGR channel order. The representation is easy to use and preserves the original camera colour information, but brightness changes affect all channels. A threshold that detects a brown-red rust patch in one exposure may fail when the image becomes darker or brighter.

RGB remains useful for display, archival and feature extraction. It is less suitable as the only colour thresholding space under the assignment's large ambient-light range. Clipping in any channel can also destroy colour information, so exposure and strobe intensity should be locked.

Engineering implications

Lock acquisition settings and use a reference sample to detect colour drift. Colour thresholds must be calibrated with the actual casing material and production lighting.

Practical validation checklist

- Capture reference samples at 0.4×, 1.0× and 1.8× illumination.
- Compare class-centroid spread in each colour space.
- Check arithmetic saturation and colour calibration drift.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

3.3 HSV hue, saturation and value

HSV separates hue from saturation and value. Hue describes the dominant colour, saturation describes colour purity and value describes brightness. A rust detector can use hue to locate brown-red candidates and use saturation to avoid assigning a meaningful hue to nearly grey highlights.

Hue is unreliable when saturation is very low. A bright metallic reflection can be almost white and therefore have unstable hue. A practical mask should require a minimum saturation and reasonable value range, then use geometry and texture to confirm a candidate. HSV is useful for fast preliminary segmentation.

Engineering implications

Lock acquisition settings and use a reference sample to detect colour drift. Colour thresholds must be calibrated with the actual casing material and production lighting.

Practical validation checklist

- Capture reference samples at 0.4×, 1.0× and 1.8× illumination.
- Compare class-centroid spread in each colour space.
- Check arithmetic saturation and colour calibration drift.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

3.4 CIE L*a*b* and Delta E

CIE L*a*b* represents lightness using L* and chromatic components using a* and b*. The a* axis broadly represents green-to-red and the b* axis blue-to-yellow. A rusted region may shift away from the healthy steel reference in these chromatic coordinates.

A simple colour distance is Delta E*ab = square root of the sum of squared differences in L*, a* and b*. The reference should be measured from a known healthy casing under the same controlled acquisition setup. More advanced Delta E formulas can be used if perceptual uniformity is important, but calibration quality remains essential.

Engineering implications

Lock acquisition settings and use a reference sample to detect colour drift. Colour thresholds must be calibrated with the actual casing material and production lighting.

Practical validation checklist

- Capture reference samples at 0.4×, 1.0× and 1.8× illumination.
- Compare class-centroid spread in each colour space.
- Check arithmetic saturation and colour calibration drift.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

3.5 Cluster dispersion and quantitative comparison

For each colour space, collect samples from healthy casing and rust regions under the three illumination conditions. Calculate the centroid of each class and the average distance of samples from that centroid. Low within-class dispersion and high between-class separation indicate a more useful representation.

The assignment does not supply measured pixel samples, so no empirical accuracy percentages can be reported. The expected engineering result is that RGB clusters shift with brightness, HSV hue is more stable when saturation is sufficient, and calibrated Lab chromatic distance provides a useful quantitative measure for rust.

Engineering implications

Lock acquisition settings and use a reference sample to detect colour drift. Colour thresholds must be calibrated with the actual casing material and production lighting.

Practical validation checklist

- Capture reference samples at 0.4×, 1.0× and 1.8× illumination.
- Compare class-centroid spread in each colour space.
- Check arithmetic saturation and colour calibration drift.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

3.6 Image arrays, channel layout and safe arithmetic

A standard OpenCV colour image is commonly represented as a NumPy array with shape (height, width, 3) and dtype uint8. The pixel at row r and column c contains B, G and R values at img[r,c,0], img[r,c,1] and img[r,c,2]. Each channel has a range from 0 to 255.

Unsigned 8-bit arithmetic can overflow. For example, 250 + 20 cannot be represented in uint8 and may wrap around. OpenCV cv2.add uses saturated arithmetic and clips the result at 255. Another safe method is to convert the image to a wider integer or floating-point type, perform the operation, clip the range and convert back.

Engineering implications

Lock acquisition settings and use a reference sample to detect colour drift. Colour thresholds must be calibrated with the actual casing material and production lighting.

Practical validation checklist

- Capture reference samples at 0.4×, 1.0× and 1.8× illumination.
- Compare class-centroid spread in each colour space.
- Check arithmetic saturation and colour calibration drift.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

3.7 Selection, calibration and drift monitoring

CIE L*a*b* with a healthy-steel reference is selected for the primary rust-distance measurement. HSV may be used as a quick candidate mask, while BGR remains the source image. The final decision should combine colour distance, connected-region size, local texture and position on the battery surface.

At the start of each shift, capture a reference sample and verify white balance, exposure and strobe intensity. If the reference colour drifts beyond a validated tolerance, raise a calibration warning. Automatic threshold adaptation without an audit record should be avoided because it can hide gradual camera or lighting faults.

Engineering implications

Lock acquisition settings and use a reference sample to detect colour drift. Colour thresholds must be calibrated with the actual casing material and production lighting.

Practical validation checklist

- Capture reference samples at 0.4×, 1.0× and 1.8× illumination.
- Compare class-centroid spread in each colour space.
- Check arithmetic saturation and colour calibration drift.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

TASK 4 – EXTENDED TECHNICAL DISCUSSION

4.1 Binary masks and neighbourhood topology

After segmentation, candidate defects are represented as foreground pixels in a binary mask. Connectivity defines which foreground pixels belong to the same object. Four-connectivity uses horizontal and vertical neighbours, while eight-connectivity also includes diagonal neighbours. The choice changes component count, area and measured scratch length.

Battery scratches can occur at any angle. A diagonal scratch may be split into many components under four-connectivity. Eight-connectivity preserves diagonal continuity but can merge objects that touch only at a corner. The segmentation rule must be selected with the expected physical defect geometry in mind.

Engineering implications

Connectivity and distance metrics affect measured defect count and size. Use calibrated physical units and test the selected rule against synthetic and real defect standards.

Practical validation checklist

- Test diagonal lines, checkerboards and noisy masks.
- Compare N4, N8 and custom m-connectivity.
- Verify 0.5 mm diameter using a calibrated target.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

4.2 Four-connectivity and its limitations

Four-connectivity treats only north, south, east and west pixels as adjacent. It is useful when corner contact should not merge two objects. It can, however, fragment a one-pixel diagonal line because successive pixels touch only at corners.

If a diagonal scratch is split into separate components, the system may count one physical scratch as several scratches. This can cause an incorrect reject decision when the rule is to reject more than two distinct scratches. The algorithm should not use component count without understanding connectivity.

Engineering implications

Connectivity and distance metrics affect measured defect count and size. Use calibrated physical units and test the selected rule against synthetic and real defect standards.

Practical validation checklist

- Test diagonal lines, checkerboards and noisy masks.
- Compare N4, N8 and custom m-connectivity.
- Verify 0.5 mm diameter using a calibrated target.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

4.3 Eight-connectivity and diagonal continuity

Eight-connectivity adds the four diagonal neighbours. It is better at preserving diagonal lines and irregular defect boundaries. For a scratch detector, this often gives a more realistic connected footprint than four-connectivity.

The disadvantage is corner ambiguity. Two separate foreground regions may become one component if they touch diagonally. Checkerboard patterns demonstrate this issue. The algorithm can reduce false merging by considering local geometry, background connectivity and m-connectivity rules.

Engineering implications

Connectivity and distance metrics affect measured defect count and size. Use calibrated physical units and test the selected rule against synthetic and real defect standards.

Practical validation checklist

- Test diagonal lines, checkerboards and noisy masks.
- Compare N4, N8 and custom m-connectivity.
- Verify 0.5 mm diameter using a calibrated target.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

4.4 Mixed m-connectivity

m-connectivity includes direct four-neighbour adjacency and allows diagonal adjacency only when the shared orthogonal neighbours do not create an ambiguous connection. It is a compromise between N4 and N8 and helps prevent certain topological loop ambiguities.

OpenCV directly supports common 4- and 8-connected component labelling, but not a one-parameter m-connectivity mode. A custom flood-fill traversal can implement the rule. The custom code should be tested at image boundaries, on diagonal lines and on checkerboard patterns before production use.

Engineering implications

Connectivity and distance metrics affect measured defect count and size. Use calibrated physical units and test the selected rule against synthetic and real defect standards.

Practical validation checklist

- Test diagonal lines, checkerboards and noisy masks.
- Compare N4, N8 and custom m-connectivity.
- Verify 0.5 mm diameter using a calibrated target.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

4.5 Synthetic tests and expected behaviour

A synthetic diagonal line should be processed under both N4 and N8. N4 is expected to split corner-only contacts; N8 should keep the line as one component. A checkerboard joint exposes diagonal ambiguity. A noisy background tests whether isolated foreground pixels can be removed by a minimum-area filter.

Record the number of components, each component's area, bounding box and orientation. Repeat the tests with small gaps and one-pixel bridges. This makes the behaviour reproducible and helps select thresholds without relying on a single photograph.

Engineering implications

Connectivity and distance metrics affect measured defect count and size. Use calibrated physical units and test the selected rule against synthetic and real defect standards.

Practical validation checklist

- Test diagonal lines, checkerboards and noisy masks.
- Compare N4, N8 and custom m-connectivity.
- Verify 0.5 mm diameter using a calibrated target.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

4.6 Distance maps and physical measurements

Euclidean distance is De = square root of dx squared plus dy squared. City-Block distance is D4 = absolute dx plus absolute dy. Chessboard distance is D8 = maximum of absolute dx and absolute dy. Euclidean distance follows straight-line geometry and is preferred for measuring physical diameters after calibration.

City-Block distance creates diamond-shaped distance contours, while Chessboard distance creates square-shaped contours. These metrics can be useful for morphology or approximate distance transforms, but they do not represent physical straight-line distance as accurately as Euclidean measurement.

Engineering implications

Connectivity and distance metrics affect measured defect count and size. Use calibrated physical units and test the selected rule against synthetic and real defect standards.

Practical validation checklist

- Test diagonal lines, checkerboards and noisy masks.
- Compare N4, N8 and custom m-connectivity.
- Verify 0.5 mm diameter using a calibrated target.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

4.7 Pixel scale, rejection rule and selection

With a 320 mm horizontal field of view, the scale is 320 divided by the number of horizontal pixels. A measured Euclidean diameter in pixels is converted to millimetres using this scale. If horizontal and vertical scales differ, use separate scale factors in the distance formula.

Eight-connectivity is selected for initial scratch continuity, with m-connectivity available to resolve ambiguous corner contacts. Reject a cell when more than two distinct scratches are confirmed or a rust region exceeds 0.5 mm equivalent diameter. Borderline regions should be marked uncertain and sent to a safe reject or quarantine path.

Engineering implications

Connectivity and distance metrics affect measured defect count and size. Use calibrated physical units and test the selected rule against synthetic and real defect standards.

Practical validation checklist

- Test diagonal lines, checkerboards and noisy masks.
- Compare N4, N8 and custom m-connectivity.
- Verify 0.5 mm diameter using a calibrated target.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

TASK 5 – EXTENDED TECHNICAL DISCUSSION

5.1 Sensor acquisition and motion

The belt moves at 1.2 metres per second. During exposure, the battery continues moving, so the captured image contains a blurred representation of the object. Motion blur reduces the visibility of small cracks, etched text and narrow rust pits. Exposure time must therefore be selected from the physical speed and allowable blur.

Motion blur distance is b = v multiplied by t. At 1.2 m/s, the object moves 1.2 mm in one millisecond. A 4 ms exposure produces 4.8 mm of blur, 1 ms produces 1.2 mm and 0.05 ms produces 0.06 mm.

Engineering implications

Exposure, shutter mode and lighting must be designed together. Verify timing using the actual sensor trigger and encoder rather than relying only on nominal specifications.

Practical validation checklist

- Measure blur at each exposure time and belt speed.
- Check rolling-shutter bands under 100 Hz flicker.
- Verify global trigger and strobe timing.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

5.2 Rolling-shutter CMOS behaviour

A rolling-shutter sensor exposes or reads different image rows at different times. A moving battery can appear skewed because one row is captured before another. This can distort dimensions and make a straight edge appear slanted.

The factory also has 100 Hz illumination flicker, corresponding to a 10 ms period. If the row readout spans a meaningful part of this period, different rows may receive different illumination levels. The image can show horizontal bands or brightness changes that resemble surface defects.

Engineering implications

Exposure, shutter mode and lighting must be designed together. Verify timing using the actual sensor trigger and encoder rather than relying only on nominal specifications.

Practical validation checklist

- Measure blur at each exposure time and belt speed.
- Check rolling-shutter bands under 100 Hz flicker.
- Verify global trigger and strobe timing.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

5.3 Global-shutter CMOS acquisition

A global-shutter sensor captures all pixels at approximately the same instant. This avoids the row-by-row geometric skew associated with rolling shutter. When combined with a short LED strobe, it can freeze the battery during motion and reduce the effect of ambient light.

The camera should support an external hardware trigger. A belt encoder provides a position reference, allowing the system to capture at defined displacement intervals. Strobe timing, exposure and camera trigger delay should be measured together rather than configured independently.

Engineering implications

Exposure, shutter mode and lighting must be designed together. Verify timing using the actual sensor trigger and encoder rather than relying only on nominal specifications.

Practical validation checklist

- Measure blur at each exposure time and belt speed.
- Check rolling-shutter bands under 100 Hz flicker.
- Verify global trigger and strobe timing.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

5.4 Line-scan acquisition

A line-scan camera captures one line of pixels at a time. The conveyor movement supplies the second image dimension as successive lines are acquired. It is useful for continuous inspection and can provide high spatial resolution along the motion direction.

Line-scan systems require encoder synchronization and a suitable line rate. If belt speed changes, line spacing changes unless the encoder controls acquisition. Object gaps, rotation and varying object height must also be handled. It is a possible premium option but adds integration complexity.

Engineering implications

Exposure, shutter mode and lighting must be designed together. Verify timing using the actual sensor trigger and encoder rather than relying only on nominal specifications.

Practical validation checklist

- Measure blur at each exposure time and belt speed.
- Check rolling-shutter bands under 100 Hz flicker.
- Verify global trigger and strobe timing.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

5.5 Motion blur table and design target

At the stated speed, exposure of 4 ms creates 4.8 mm blur, 1 ms creates 1.2 mm blur and 0.05 ms creates 0.06 mm blur. At Full HD scale of approximately 0.1667 mm per pixel, these correspond to about 28.8, 7.2 and 0.36 pixels. At a 4096-pixel width, they correspond to about 61.5, 15.4 and 0.77 pixels.

For a 0.3 mm defect, a useful initial design target is to keep motion blur well below the defect dimension. A 0.05 ms strobe exposure is much more appropriate than 1–4 ms, but the final setting must be checked against sensor sensitivity and actual image signal-to-noise ratio.

Engineering implications

Exposure, shutter mode and lighting must be designed together. Verify timing using the actual sensor trigger and encoder rather than relying only on nominal specifications.

Practical validation checklist

- Measure blur at each exposure time and belt speed.
- Check rolling-shutter bands under 100 Hz flicker.
- Verify global trigger and strobe timing.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

5.6 Photon-to-digital signal chain

Photons reach the sensor through the lens and generate charge in photosensitive pixels. The sensor integrates charge during the exposure period. Readout circuitry converts charge to voltage, analog conditioning amplifies and filters the signal, and an analog-to-digital converter quantizes the voltage into digital pixel values.

Noise may enter through photon statistics, dark current, readout circuitry, amplification and quantization. Stable illumination, appropriate exposure, sensor cooling where applicable and calibration improve image quality. A short exposure should be paired with sufficient strobe energy so that the image is not underexposed.

Engineering implications

Exposure, shutter mode and lighting must be designed together. Verify timing using the actual sensor trigger and encoder rather than relying only on nominal specifications.

Practical validation checklist

- Measure blur at each exposure time and belt speed.
- Check rolling-shutter bands under 100 Hz flicker.
- Verify global trigger and strobe timing.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

5.7 CCD versus CMOS and final selection

CCD sensors transfer charge through the device to an output node. CMOS sensors integrate readout circuitry near the pixels and commonly support high frame rates, lower power and flexible trigger modes. Modern industrial CMOS devices are available with global shutters.

A global-shutter CMOS camera with synchronized LED strobe is selected for the baseline line. It best addresses moving objects, ambient flicker and trigger control. Final procurement should compare actual read noise, dynamic range, shutter efficiency, frame rate, trigger delay and interface bandwidth.

Engineering implications

Exposure, shutter mode and lighting must be designed together. Verify timing using the actual sensor trigger and encoder rather than relying only on nominal specifications.

Practical validation checklist

- Measure blur at each exposure time and belt speed.
- Check rolling-shutter bands under 100 Hz flicker.
- Verify global trigger and strobe timing.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

TASK 6 – EXTENDED TECHNICAL DISCUSSION

6.1 Spatial sampling and field of view

The horizontal field of view is 320 mm. If the camera provides W horizontal pixels, the sampling scale is 320/W millimetres per pixel. A smaller scale means that a physical defect occupies more pixels. The 0.3 mm micro-crack and 0.4 mm text stroke are the critical details in the assignment.

Sampling is not the same as optical resolution. The lens, focus, sensor modulation transfer function, motion blur and contrast all affect whether the pixel samples contain usable detail. A theoretical pixel count alone cannot guarantee successful defect detection.

Engineering implications

Resolution selection must include optical quality, bandwidth and storage. A pixel count that is adequate mathematically may still fail if the lens or motion blur removes detail.

Practical validation checklist

- Measure the actual mm/px scale with a target.
- Calculate interface data rate at the required frame rate.
- Confirm five-day evidence retention within storage limits.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

6.2 VGA resolution assessment

VGA provides 640 horizontal pixels. The scale is 320/640 = 0.5 mm per pixel. A 0.3 mm defect occupies only 0.6 pixel and a 0.4 mm text stroke occupies 0.8 pixel. Both are below reliable sampling.

VGA has low bandwidth and storage requirements, but it is suitable only for coarse object presence or large shape differences. It cannot be justified for the smallest specified defects across the full 320 mm field of view.

Engineering implications

Resolution selection must include optical quality, bandwidth and storage. A pixel count that is adequate mathematically may still fail if the lens or motion blur removes detail.

Practical validation checklist

- Measure the actual mm/px scale with a target.
- Calculate interface data rate at the required frame rate.
- Confirm five-day evidence retention within storage limits.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

6.3 Full HD resolution assessment

Full HD provides 1920 horizontal pixels. The scale is 320/1920 = 0.1667 mm per pixel. A 0.3 mm defect spans about 1.8 pixels and a 0.4 mm text stroke spans about 2.4 pixels. This is better than VGA but remains marginal for robust measurement.

Full HD is a practical baseline because it reduces CPU and bandwidth load compared with 12 MP. However, the system should be tested with real defect standards. If a 0.3 mm crack must be detected consistently, narrowing the field of view or using a higher-resolution sensor may be necessary.

Engineering implications

Resolution selection must include optical quality, bandwidth and storage. A pixel count that is adequate mathematically may still fail if the lens or motion blur removes detail.

Practical validation checklist

- Measure the actual mm/px scale with a target.
- Calculate interface data rate at the required frame rate.
- Confirm five-day evidence retention within storage limits.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

6.4 12 MP resolution assessment

A 4096-pixel horizontal image across 320 mm gives 0.078125 mm per pixel. A 0.3 mm defect spans about 3.84 pixels and a 0.4 mm stroke spans about 5.12 pixels. This provides a more useful sampling margin for small surface features.

The trade-off is data volume. A 12 MP RGB frame contains approximately 36.9 million bytes when stored uncompressed at three bytes per pixel. At 15 frames per second, the data rate is roughly 553 MB/s, which is far above the practical payload of Gigabit Ethernet.

Engineering implications

Resolution selection must include optical quality, bandwidth and storage. A pixel count that is adequate mathematically may still fail if the lens or motion blur removes detail.

Practical validation checklist

- Measure the actual mm/px scale with a target.
- Calculate interface data rate at the required frame rate.
- Confirm five-day evidence retention within storage limits.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

6.5 Nyquist principle and practical sampling

The Nyquist principle states that a band-limited signal must be sampled at more than twice its highest spatial frequency to avoid aliasing. For a 0.4 mm feature period, the theoretical minimum is two samples per feature, or 5 samples per millimetre. Across 320 mm this corresponds to 1600 horizontal samples.

Practical inspection usually requires more than two samples across a feature because edges are blurred by optics and motion, and because classification needs shape rather than mere detection. The 12 MP option gives more margin, while Full HD is near the lower practical boundary for the smallest text stroke.

Engineering implications

Resolution selection must include optical quality, bandwidth and storage. A pixel count that is adequate mathematically may still fail if the lens or motion blur removes detail.

Practical validation checklist

- Measure the actual mm/px scale with a target.
- Calculate interface data rate at the required frame rate.
- Confirm five-day evidence retention within storage limits.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

6.6 Bandwidth, storage and five-day archive

Five days contain 432,000 seconds. At 15 cells per second, 6,480,000 images are produced if one image is retained for every cell. With uncompressed RGB, the approximate archive is 5.97 TB for VGA, 40.31 TB for Full HD and 238.88 TB for 12 MP. These values exceed the 1.5 TB SSD capacity.

The archive requirement therefore needs a storage policy. Options include lossless compression, cropping to a validated ROI, storing compact proof images for all cells and retaining higher-detail raw images for flagged cases. Any policy must preserve sufficient evidence for compliance audits.

Engineering implications

Resolution selection must include optical quality, bandwidth and storage. A pixel count that is adequate mathematically may still fail if the lens or motion blur removes detail.

Practical validation checklist

- Measure the actual mm/px scale with a target.
- Calculate interface data rate at the required frame rate.
- Confirm five-day evidence retention within storage limits.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

6.7 Frame rate, object coverage and selection

The 70 mm cell must be captured within the 240 mm vertical field of view. The frame rate must be coordinated with object speed and the displacement between frames. Encoder-triggered acquisition and overlapping views are safer than relying on an arbitrary fixed frame interval, especially if cell spacing varies.

Full HD is the balanced baseline for throughput, but its sampling of a 0.3 mm defect is marginal. A high-resolution camera or narrower ROI should be considered if tests show that Full HD cannot meet the required miss rate. Storage should be controlled through a validated evidence strategy rather than reducing spatial resolution below the safety need.

Engineering implications

Resolution selection must include optical quality, bandwidth and storage. A pixel count that is adequate mathematically may still fail if the lens or motion blur removes detail.

Practical validation checklist

- Measure the actual mm/px scale with a target.
- Calculate interface data rate at the required frame rate.
- Confirm five-day evidence retention within storage limits.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

TASK 7 – EXTENDED TECHNICAL DISCUSSION

7.1 Quantization and bit-depth fundamentals

An n-bit image represents 2 to the power n discrete intensity levels. Four-bit data has 16 levels, six-bit has 64, eight-bit has 256 and twelve-bit has 4096. The quantization step depends on the signal range and gain. If the step is larger than the defect-induced intensity change, the defect may disappear.

The assignment states that a dent or deformity can create only a 5–8 grey-level drop on an 8-bit scale. Therefore, the camera must preserve subtle intensity differences and avoid clipping or excessive noise.

Engineering implications

Preserve enough intensity precision to see the subtle defect contrast. Do not confuse camera bit depth with neural-network arithmetic precision.

Practical validation checklist

- Compare 4-, 6-, 8- and 12-bit images of the same sample.
- Measure subtle 5–8 level contrast after acquisition.
- Check banding, noise and file size.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

7.2 Four-bit and six-bit acquisition

When the full 0–255 range is mapped to 16 levels, each 4-bit step is approximately 17 intensity values. A 5–8 level change can be lost completely. Six-bit acquisition has 64 levels and a step of about 4 values, so the same change may survive as only one or two steps.

Four-bit acquisition is not suitable for subtle dent inspection. Six-bit acquisition may reduce storage but leaves little margin for noise, lighting variation and compression. It should only be used if tests demonstrate that the target defect remains reliably visible.

Engineering implications

Preserve enough intensity precision to see the subtle defect contrast. Do not confuse camera bit depth with neural-network arithmetic precision.

Practical validation checklist

- Compare 4-, 6-, 8- and 12-bit images of the same sample.
- Measure subtle 5–8 level contrast after acquisition.
- Check banding, noise and file size.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

7.3 Eight-bit industrial acquisition

Eight-bit acquisition provides 256 intensity values and preserves a 5–8 level difference as multiple code values. It is a practical industrial standard for many vision applications and keeps memory use manageable.

Eight-bit precision is not enough by itself. The sensor must have adequate signal-to-noise ratio, exposure must be stable and highlights must not saturate. A diffuse strobe enclosure reduces variation caused by the factory's changing ambient light.

Engineering implications

Preserve enough intensity precision to see the subtle defect contrast. Do not confuse camera bit depth with neural-network arithmetic precision.

Practical validation checklist

- Compare 4-, 6-, 8- and 12-bit images of the same sample.
- Measure subtle 5–8 level contrast after acquisition.
- Check banding, noise and file size.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

7.4 Twelve-bit raw acquisition

Twelve-bit raw acquisition provides 4096 levels and can preserve finer signal differences if the sensor's analog performance supports them. It offers more headroom for bright and dark regions and can help quantitative photometric analysis.

More digital levels do not automatically improve the image if sensor noise is larger than the subtle defect signal. The effective number of bits, full-well capacity and read noise should be checked. Twelve-bit data also increases storage and processing requirements.

Engineering implications

Preserve enough intensity precision to see the subtle defect contrast. Do not confuse camera bit depth with neural-network arithmetic precision.

Practical validation checklist

- Compare 4-, 6-, 8- and 12-bit images of the same sample.
- Measure subtle 5–8 level contrast after acquisition.
- Check banding, noise and file size.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

7.5 False contouring and smooth surface gradients

Down-quantization converts a smooth brightness gradient into a smaller number of constant-value bands. These bands can create false contours over a curved battery shoulder and hide the gradual intensity change caused by a dent.

A useful experiment is to capture a smooth curved surface and quantize the same image to 4, 6, 8 and 12 bits. Compare the number of unique levels, gradient maps and edge maps. The 4-bit result should show the strongest banding, while 8- and 12-bit results preserve more subtle gradation.

Engineering implications

Preserve enough intensity precision to see the subtle defect contrast. Do not confuse camera bit depth with neural-network arithmetic precision.

Practical validation checklist

- Compare 4-, 6-, 8- and 12-bit images of the same sample.
- Measure subtle 5–8 level contrast after acquisition.
- Check banding, noise and file size.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

7.6 Image bit depth versus model quantization

Image bit depth describes the digitization of sensor intensity. Model quantization describes how model weights and activations are represented during inference. FP32 uses 32-bit floating point, while INT8 uses 8-bit integer representation on supported hardware.

These are separate decisions. The camera may capture an 8-bit or 12-bit image while a machine-learning model uses INT8 arithmetic. Lowering model precision can reduce memory and inference cost, but it must be checked for classification accuracy and must not be confused with lowering the sensor's image bit depth.

Engineering implications

Preserve enough intensity precision to see the subtle defect contrast. Do not confuse camera bit depth with neural-network arithmetic precision.

Practical validation checklist

- Compare 4-, 6-, 8- and 12-bit images of the same sample.
- Measure subtle 5–8 level contrast after acquisition.
- Check banding, noise and file size.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

7.7 Balanced choice and validation

Eight-bit industrial acquisition is selected as the baseline because it preserves the specified 5–8 level contrast while limiting data volume. Twelve-bit capture can be evaluated for difficult cases if the camera supports it at the required frame rate. Four-bit is rejected for the subtle defect requirement.

Test each bit depth using identical samples, illumination and exposure. Measure defect recall, false-reject rate and processing time. The final selection should preserve the smallest required defect without exceeding storage and compute limits.

Engineering implications

Preserve enough intensity precision to see the subtle defect contrast. Do not confuse camera bit depth with neural-network arithmetic precision.

Practical validation checklist

- Compare 4-, 6-, 8- and 12-bit images of the same sample.
- Measure subtle 5–8 level contrast after acquisition.
- Check banding, noise and file size.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

TASK 8 – EXTENDED TECHNICAL DISCUSSION

8.1 Image formation and reflective surfaces

A digital image is formed when light reflected from the battery passes through the lens and reaches the sensor. The final pixel values depend on illumination, object reflectance, viewing geometry, lens transmission, sensor response and exposure. Reflective metal can produce bright specular highlights that conceal scratches or resemble dents.

The line therefore needs controlled diffuse illumination. A light dome or tunnel reduces direct reflections and makes the surface appearance more repeatable. Exposure and camera gain should be locked after calibration so that the image does not change unpredictably with ambient light.

Engineering implications

Calibration must cover both geometry and illumination. A belt-plane calibration alone may not correct measurements on a raised battery surface.

Practical validation checklist

- Calibrate focal length and lens distortion.
- Measure dimensional error at belt and battery-top heights.
- Capture flat-field and dark-reference frames.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

8.2 Standard lens with ambient light

A standard fixed-focal-length lens is economical and easy to install. However, ambient light varies from 1,500 to 45,000 lux, which is a large change in illumination. Automatic exposure may compensate by changing brightness but can also change the appearance of defects from one frame to another.

Perspective projection causes nearer parts of an object to appear larger. A scale calibrated at the conveyor plane may not be accurate for the top surface of a battery that is 30 mm above the belt. This matters when the system measures pouch swelling or dimensions.

Engineering implications

Calibration must cover both geometry and illumination. A belt-plane calibration alone may not correct measurements on a raised battery surface.

Practical validation checklist

- Calibrate focal length and lens distortion.
- Measure dimensional error at belt and battery-top heights.
- Capture flat-field and dark-reference frames.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

8.3 Diffuse dome lighting and checkerboard calibration

A diffuse LED dome surrounds the object with a broad, even light field. It reduces sharp reflections and helps illuminate curved metallic jackets. A checkerboard target provides known geometric points for estimating camera intrinsics and lens distortion.

Calibration should estimate focal length, principal point and distortion coefficients. Undistortion corrects lens distortion, while a planar homography can map the belt plane to a top-down coordinate system. A planar correction alone does not remove all height-dependent perspective error.

Engineering implications

Calibration must cover both geometry and illumination. A belt-plane calibration alone may not correct measurements on a raised battery surface.

Practical validation checklist

- Calibrate focal length and lens distortion.
- Measure dimensional error at belt and battery-top heights.
- Capture flat-field and dark-reference frames.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

8.4 Telecentric lens and dimensional accuracy

An object-space telecentric lens makes chief rays approximately parallel in object space. It reduces magnification changes when the object moves closer to or farther from the lens. This is valuable for dimensional measurement and can improve accuracy for raised battery surfaces.

Telecentric optics are more expensive and may have a limited field of view. Under the fixed budget, a calibrated standard lens with diffuse illumination is the practical baseline. A telecentric lens should be considered if measured dimensional errors remain outside the ±0.5 mm requirement.

Engineering implications

Calibration must cover both geometry and illumination. A belt-plane calibration alone may not correct measurements on a raised battery surface.

Practical validation checklist

- Calibrate focal length and lens distortion.
- Measure dimensional error at belt and battery-top heights.
- Capture flat-field and dark-reference frames.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

8.5 Focal length estimate

Using the thin-lens field-of-view approximation, focal length is approximately sensor width multiplied by working distance divided by required field of view. For an approximate 1/2-inch sensor width of 6.4 mm, a 400 mm working distance and 320 mm horizontal field of view give $f \approx 6.4 \times 400 / 320 = 8$ mm.

This is an initial estimate. The actual sensor active width, lens distortion, focus distance and camera mount must be checked against the lens datasheet. The final field of view should be verified with a calibrated target at the belt plane.

Engineering implications

Calibration must cover both geometry and illumination. A belt-plane calibration alone may not correct measurements on a raised battery surface.

Practical validation checklist

- Calibrate focal length and lens distortion.
- Measure dimensional error at belt and battery-top heights.
- Capture flat-field and dark-reference frames.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

8.6 Perspective error for a 30 mm high object

At the belt plane, the camera distance is 400 mm. At the top of a 30 mm high battery, the distance is 370 mm. Since projected image size is proportional to 1/Z, the top appears larger by a ratio of 400/370 = 1.0811.

The relative apparent scale error is approximately 8.11%. This is large compared with a ±0.5 mm dimensional tolerance. The design should measure at a known surface plane, use a height-aware correction, or adopt telecentric optics if the tolerance cannot otherwise be met.

Engineering implications

Calibration must cover both geometry and illumination. A belt-plane calibration alone may not correct measurements on a raised battery surface.

Practical validation checklist

- Calibrate focal length and lens distortion.
- Measure dimensional error at belt and battery-top heights.
- Capture flat-field and dark-reference frames.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

8.7 Flat-field correction and homomorphic filtering

Flat-field correction uses an image of a uniform target to estimate spatial illumination variation. A dark frame estimates sensor offset. The corrected image is calculated from the input minus the dark frame, divided by the flat-field response minus the dark frame, then scaled to a convenient mean intensity.

Homomorphic filtering models the image as the product of illumination and reflectance. Taking logarithms changes the product into a sum, log(f) = log(i) + log(r). A low-frequency estimate of illumination can then be removed to emphasize reflectance detail. These methods complement controlled lighting but cannot restore clipped highlights.

Engineering implications

Calibration must cover both geometry and illumination. A belt-plane calibration alone may not correct measurements on a raised battery surface.

Practical validation checklist

- Calibrate focal length and lens distortion.
- Measure dimensional error at belt and battery-top heights.
- Capture flat-field and dark-reference frames.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

TASK 9 – EXTENDED TECHNICAL DISCUSSION

9.1 Integration requirements and architecture

The integrated system must connect optics, camera, illumination, image processing, classification, object tracking, PLC output and evidence storage. Each stage has a timing budget and a failure mode. The system should be designed around the 400 ms physical travel window and the 15 cells per second arrival rate.

The object tracker should use encoder position and a unique cell identifier. Every image and decision must remain associated with the correct battery until the reject gate. A lost frame or uncertain track should trigger a safe reject or quarantine action.

Engineering implications

The integrated design must meet timing, budget, storage and audit requirements together. Test the complete camera-to-piston loop, not only the image algorithm.

Practical validation checklist

- Time every processing and control stage.
- Test under changing light, vibration and glare.
- Run endurance tests at the full line rate.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

9.2 Budget configuration and its limitations

The budget configuration uses a rolling-shutter USB camera, ambient lighting, JPEG storage and a low-plus-mid-level processing stack. It has a low initial cost but is vulnerable to motion skew, flicker, glare and compression artefacts.

JPEG is lossy and may alter fine edges or small texture patterns. Ambient lighting changes can shift image intensity. This package can be used for demonstration or coarse sorting but is not the preferred safety inspection architecture.

Engineering implications

The integrated design must meet timing, budget, storage and audit requirements together. Test the complete camera-to-piston loop, not only the image algorithm.

Practical validation checklist

- Time every processing and control stage.
- Test under changing light, vibration and glare.
- Run endurance tests at the full line rate.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

9.3 Balanced configuration

The balanced configuration uses a global-shutter GigE camera, synchronized diffuse LED dome or strobe, calibrated standard lens, 8-bit lossless evidence and a three-level processing stack. It provides controlled acquisition and interpretable classification while remaining compatible with a CPU-based edge computer.

The main limitation is spatial sampling at 2 MP across a 320 mm field of view. A 0.3 mm defect spans about 1.8 pixels, so the optical and defect performance must be tested. A narrower ROI or higher-resolution camera may be needed if the required defect recall is not achieved.

Engineering implications

The integrated design must meet timing, budget, storage and audit requirements together. Test the complete camera-to-piston loop, not only the image algorithm.

Practical validation checklist

- Time every processing and control stage.
- Test under changing light, vibration and glare.
- Run endurance tests at the full line rate.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

9.4 Premium configuration

The premium configuration uses a 4k line-scan camera, telecentric lens, uniform line illumination and 12-bit raw storage. It can provide high spatial detail and continuous coverage, but it requires accurate encoder synchronization and higher data throughput.

The premium configuration is more expensive and can exceed the CPU, interface and storage limits. It may be justified for a later high-precision station, but it is not the best first deployment under the original budget.

Engineering implications

The integrated design must meet timing, budget, storage and audit requirements together. Test the complete camera-to-piston loop, not only the image algorithm.

Practical validation checklist

- Time every processing and control stage.
- Test under changing light, vibration and glare.
- Run endurance tests at the full line rate.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

9.5 Indicative cost allocation and procurement

A balanced planning allowance can be divided among the global-shutter camera, lens, diffuse illumination, strobe controller, encoder interface, mount and enclosure, cables, edge computer and commissioning. These figures are budget targets only; current vendor quotations and product compatibility must be confirmed before purchasing.

Procurement should check camera frame rate at the selected resolution, global-shutter availability, trigger input/output, GigE bandwidth, lens sensor format, illumination power, enclosure dimensions and software driver support. The system should not be approved based only on a camera's headline megapixel count.

Engineering implications

The integrated design must meet timing, budget, storage and audit requirements together. Test the complete camera-to-piston loop, not only the image algorithm.

Practical validation checklist

- Time every processing and control stage.
- Test under changing light, vibration and glare.
- Run endurance tests at the full line rate.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

9.6 Timing allocation and control margin

The 0.48 m camera-to-piston distance at 1.2 m/s gives 400 ms of travel time. A design budget should allocate time to exposure, sensor readout, frame transfer, preprocessing, segmentation, OCR, classification, PLC communication and reserve margin.

The software target should be comfortably below 400 ms. Average latency alone is not sufficient; 95th- or 99th-percentile latency should be measured under worst-case images. The controller must account for communication jitter and actuator response.

Engineering implications

The integrated design must meet timing, budget, storage and audit requirements together. Test the complete camera-to-piston loop, not only the image algorithm.

Practical validation checklist

- Time every processing and control stage.
- Test under changing light, vibration and glare.
- Run endurance tests at the full line rate.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

9.7 Decision matrix and commissioning

The balanced package is selected for the original line because it offers a strong compromise between robustness, cost, compute and maintainability. The diffuse strobe enclosure addresses storm-cloud illumination; global shutter and short exposure address motion; calibration and confidence rules address measurement and classification uncertainty.

Commissioning should verify image quality, illumination uniformity, encoder tracking, actuator timing, archive retrieval and per-class defect performance. The system should demonstrate the required ≤1.5% miss rate on an independent representative dataset before release.

Engineering implications

The integrated design must meet timing, budget, storage and audit requirements together. Test the complete camera-to-piston loop, not only the image algorithm.

Practical validation checklist

- Time every processing and control stage.
- Test under changing light, vibration and glare.
- Run endurance tests at the full line rate.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

TASK 10 – EXTENDED TECHNICAL DISCUSSION

10.1 Scale-up requirements and architecture change

The new operating point doubles belt speed to 2.4 m/s and throughput to 30 cells per second. The decision window is reduced to 200 ms. In addition, the client requires detection of internal short-circuit indicators through thermal spots with variations as small as 0.5°C in the MWIR spectrum.

This is not merely a faster version of the original visible inspection task. The new requirement introduces a different sensing modality. Visible cameras cannot directly detect a thermal feature that is invisible in visible light, so the architecture must include thermal sensing.

Engineering implications

The scale-up introduces new speed and spectral requirements. A visible-only redesign cannot satisfy a thermal detection requirement; synchronized MWIR sensing is necessary.

Practical validation checklist

- Verify 30 cells/s and 200 ms decision timing.
- Calibrate MWIR response around a 0.5°C difference.
- Confirm visible/thermal registration and safe fallback.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

10.2 Strategy A – retain and optimize

Retaining the existing camera and optimizing regions of interest can reduce processing time. Shorter exposure can reduce motion blur, and downsampling unimportant image regions can lower CPU load. This strategy has the lowest cost and may help meet the higher visible inspection rate.

However, it cannot satisfy the MWIR thermal requirement. It may also fail if the existing camera cannot sustain 30 cells per second or if the interface bandwidth is insufficient. It is therefore not adequate as the complete redesign.

Engineering implications

The scale-up introduces new speed and spectral requirements. A visible-only redesign cannot satisfy a thermal detection requirement; synchronized MWIR sensing is necessary.

Practical validation checklist

- Verify 30 cells/s and 200 ms decision timing.
- Calibrate MWIR response around a 0.5°C difference.
- Confirm visible/thermal registration and safe fallback.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

10.3 Strategy B – modify sensing and shutter

An upgraded global-shutter camera, shorter synchronized strobe, enclosed light tunnel and optimized processing can improve visible acquisition at the higher speed. INT8 model inference may reduce compute cost when the processor supports efficient integer operations.

INT8 model quantization is a software optimization. It does not turn a visible camera into a thermal camera. Strategy B improves visible inspection but still cannot detect internal thermal anomalies without an MWIR sensor.

Engineering implications

The scale-up introduces new speed and spectral requirements. A visible-only redesign cannot satisfy a thermal detection requirement; synchronized MWIR sensing is necessary.

Practical validation checklist

- Verify 30 cells/s and 200 ms decision timing.
- Calibrate MWIR response around a 0.5°C difference.
- Confirm visible/thermal registration and safe fallback.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

10.4 Strategy C – dual-stream visible and MWIR

A dual-stream architecture uses a high-speed visible sensor for form factor, chemistry and surface defects, plus a synchronized MWIR thermal sensor for hot-spot detection. The sensors must share a timing and position reference so that both observations correspond to the same cell.

The thermal sensor should operate in a controlled enclosure and be calibrated using suitable reference sources. A thermal hot spot is an indicator, not automatic proof of an internal short circuit. Decision rules must be validated with representative cells and safe test procedures.

Engineering implications

The scale-up introduces new speed and spectral requirements. A visible-only redesign cannot satisfy a thermal detection requirement; synchronized MWIR sensing is necessary.

Practical validation checklist

- Verify 30 cells/s and 200 ms decision timing.
- Calibrate MWIR response around a 0.5°C difference.
- Confirm visible/thermal registration and safe fallback.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

10.5 Motion blur and exposure at 2.4 m/s

At 2.4 m/s, the belt moves 2.4 mm per millisecond. Therefore, 4 ms produces 9.6 mm blur, 1 ms produces 2.4 mm and 0.05 ms produces 0.12 mm. To limit blur to 0.1 mm, the exposure should be no longer than about 41.7 microseconds.

The strobe should be synchronized to the camera and encoder. The sensor must have adequate sensitivity to capture useful detail at this short exposure. The actual blur should be measured using a moving target at maximum speed.

Engineering implications

The scale-up introduces new speed and spectral requirements. A visible-only redesign cannot satisfy a thermal detection requirement; synchronized MWIR sensing is necessary.

Practical validation checklist

- Verify 30 cells/s and 200 ms decision timing.
- Calibrate MWIR response around a 0.5°C difference.
- Confirm visible/thermal registration and safe fallback.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

10.6 Throughput and data-rate consequences

At 30 cells per second, the average arrival interval is 33.3 ms. A pipelined system must process 30 new cells every second and associate each result with its object. At Full HD RGB, 6.22 MB per frame multiplied by 30 gives about 186.6 MB/s. At 12 MP RGB, the rate is about 1.1 GB/s.

These data rates exceed practical Gigabit Ethernet capacity for uncompressed full-frame transfer. The design may require ROI transfer, compression, a faster interface or separate processing paths. The archive plan must also be revised because the number of images per day doubles.

Engineering implications

The scale-up introduces new speed and spectral requirements. A visible-only redesign cannot satisfy a thermal detection requirement; synchronized MWIR sensing is necessary.

Practical validation checklist

- Verify 30 cells/s and 200 ms decision timing.
- Calibrate MWIR response around a 0.5°C difference.
- Confirm visible/thermal registration and safe fallback.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

10.7 Thermal calibration, fusion and final pivot

MWIR sensing must resolve the required 0.5°C variation under the actual operating conditions. Emissivity, reflected radiation, sensor drift and enclosure temperature can affect measurements. A reference source and controlled measurement geometry are needed.

The recommended pivot is REDESIGN for the complete scale-up requirement. Retain the existing visible inspection concept where useful, upgrade acquisition for speed, and add a synchronized MWIR stream. Use decision-level fusion, conservative confidence thresholds and a quarantine action when either sensor stream is missing or misaligned.

Engineering implications

The scale-up introduces new speed and spectral requirements. A visible-only redesign cannot satisfy a thermal detection requirement; synchronized MWIR sensing is necessary.

Practical validation checklist

- Verify 30 cells/s and 200 ms decision timing.
- Calibrate MWIR response around a 0.5°C difference.
- Confirm visible/thermal registration and safe fallback.

Evidence item	Observation / measured value	Status
Test ID / sample ID	To be completed during practical validation	Pending

APPENDIX – CALCULATION CHECKS AND ACCEPTANCE RECORD

Core calculations

Calculation	Formula	Result
Cell arrival interval at 15 cells/s	1 / 15	66.7 ms
Camera-to-piston travel time	0.48 m / 1.2 m/s	400 ms
Motion blur at 1.2 m/s, 4 ms	1.2 × 0.004	4.8 mm
Motion blur at 1.2 m/s, 1 ms	1.2 × 0.001	1.2 mm
Motion blur at 1.2 m/s, 0.05 ms	1.2 × 0.00005	0.06 mm
Full HD horizontal scale	320 / 1920	0.1667 mm/px
12 MP horizontal scale	320 / 4096	0.078125 mm/px
Focal length estimate	(6.4 × 400) / 320	8 mm
Perspective magnification ratio	400 / 370	1.0811
Perspective size error	(1.0811 - 1) × 100	8.11%
Scale-up blur at 2.4 m/s, 0.05 ms	2.4 × 0.00005	0.12 mm
Scale-up exposure for 0.1 mm blur	0.1 / 2400	41.7 µs

Data and measurement limitations

The assignment brief provides the operating constraints but does not provide measured camera benchmark results, a labelled ground-truth image set, vendor quotations or a measured MWIR thermal dataset. Calculations derived directly from the supplied values are presented as calculations. Accuracy, latency, supplier cost and thermal detection performance must be measured or confirmed before deployment.

The extended discussion is intended to provide a detailed engineering submission. Practical evidence fields are left open for actual test measurements; they should be completed only after the relevant experiment has been performed.